SR-HC series

Industrial Robot

Robot controller TS3000
Robot controller TS3000E
Robot controller TS3100
Robot controller TS3100E

INSTRUCTION MANUAL

WAFER TRANSFER ROBOT

MAINTENANCE MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Rev.5

October 20, 2022

Shibaura Machine

This Instruction Manual applies to the following robots:

SR Series: SR-404HC, SR-405HC-L1550-R/L, SR-405HC-L1530S-R/L, SR-624HC

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WARRANTY

This product is delivered to each customer only after it is inspected very carefully to make sure that it satisfies the SHIBAURA MACHINE's standard. Should it cause an inconvenience, we will guarantee as described below.

1. Warranty period

   SHIBAURA MACHINE agrees to repair or replace as necessary all defective material or workmanship up to the period shown below, whichever comes first.

   1) Twelve (12) months from the date of dispatch from our plant.
   2) 2,000 running hours from the date of initial machine operation.

2. Contents of warranty

   1) Only the product is subject to SHIBAURA MACHINE's Guarantee. Such Guarantee covers the specifications and functions as defined in the product specifications manual, catalog, instruction manual, etc. In no event does the Guarantee cover any secondary or associated damage caused by malfunction of the product.
   2) SHIBAURA MACHINE repairs the product free of charge only when it malfunctioned after handling or use according to the instruction manual attached to the product within the specified warranty period.

3. Exemption from responsibility

   SHIBAURA MACHINE's Guarantee shall not cover the following cases.

   1) Incorrect use not described in the instruction manual, and trouble or damage caused by negligent use.
   2) Inconvenience caused by aged deterioration or long-term usage (natural fading of coating or painting, deterioration of consumable parts (*1), etc.).
   3) Inconvenience caused by sensuous phenomena (noise generation, etc. which will not affect the function).
   4) Remodeling or disassembly which SHIBAURA MACHINE will not permit.
   5) Trouble and damage caused by insufficient maintenance/inspection or improper repair.
   6) Trouble and damage caused by disaster, fire or other external factor.
   7) Internal data such as program and point which were created by the customer.
   8) When the product purchased in Japan was shipped overseas.

4. Precautions

1) Unless the product was used pursuant to its specifications, SHIBAURA MACHINE will not guarantee the basic performance of the product.

2) If the customer did not observe the warnings and cautions described in this manual, SHIBAURA MACHINE will not assume the responsibility for any consequential accident resulting in injury or death, damage or trouble.

3) Please note that the warnings, cautions and other descriptions stipulated in this manual are only those which can be assumed by SHIBAURA MACHINE as of now.

(*1) The consumable parts signify the replacement parts for maintenance as listed in “Section 5, Replacement Parts for Maintenance" of this manual.

INTRODUCTION

This manual describes the maintenance of this product, SR-HC series, and the TS3000, TS3000E, TS3100 and TS3100E controllers.

The maintenance and inspection are essential to maintain the product performance for long years to prevent a trouble and improve the safe work.
Before starting an actual operation, it is strongly recommended to read through this manual and draw up a maintenance schedule.

CAUTIONS ON SAFETY

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that “incorrect handling will imminently lead to fatalities or serious injuries.”
 **WARNING**	This means that “incorrect handling may lead to fatalities or serious injuries.”
 **CAUTION**	This means that “incorrect handling may lead to personal injuries*1) or physical damage*2).”

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger and caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

[Maintenance and inspection]

Be sure to observe the following items to use this product safely.

 **DANGER**	
 Prohibited	• DO NOT incinerate, disassemble or charge the batteries. Otherwise, they may rupture.
 Mandatory	• Be sure to remove the power plug of the controller from the power source before starting inspection or maintenance. • Batteries should be disposed of according to the user's in-house regulations.

 **CAUTION**	
 Disassembly prohibited	• The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.
 Mandatory	• Always use the SHIBAURA MACHINE's designated spare parts when replacing the parts. • Maintenance and inspection should be performed regularly. Otherwise, the system may malfunction or accidents will be caused.

Table of Contents

1. Maintenance 9
1.1 Maintenance Schedule 9
1.2 Items for Maintenance and Inspection 10
1.2.1 Inspection at Power OFF (at Non-Operation) 10
1.2.2 Inspection at Power ON (at Operation) 11
1.2.3 Overhaul 12
1.3 Cautions on Maintenance and Inspection 13
1.4 Maintenance Tools 13
1.5 Clamping Hexagon Socket Head Cap Screws and Set screws 14
1.6 Maintenance Contract and Repair 15
1.6.1 Maintenance Contract 15
1.6.2 Repair 15
1.6.3 Modification 16

2. Maintenance of the Main Robot 17
2.1 Layout of Robot Components and Drive Mechanism 17
2.2 Details of Inspection 19
2.2.1 Check of Each Bolt (or Screw) for Clamping 19
2.2.2 Check of Each Connector for Clamping 21
2.2.3 Check for Vibration and Abnormal Noise 22
2.2.4 Check for Lubrication 23
2.2.5 Check for Wiring 26
2.2.6 Check for Timing Belt Status 27
2.2.7 Check for Resin Belt 27
2.3 Replacing Position Detector (or Encoder) Batteries 28
2.3.1 Replacing Batteries 29
2.3.2 Battery Error Code 30

3 Maintenance of Controller 32
3.1 Cautions on Maintenance and Inspection 32
3.2 Layout of Controller Parts (TS3000, TS3000E) 33
3.3 Layout of Controller Parts (TS3100, TS3100E) 34
3.4 Maintenance Procedures 35
3.4.1 Check for Controller Air Vent Holes 35
3.4.2 Check of Safety Devices for Function 37
3.4.3 Battery Replacement 40
3.4.4 Replacement of Switching Power Supply Unit 42
3.4.5 Replacement of Fuse (X8GN, X8GI Printed Board) 48
3.4.6 Replacement of Output ICs (X8GC and X8GN (X8GI) Printed Boards) . 49

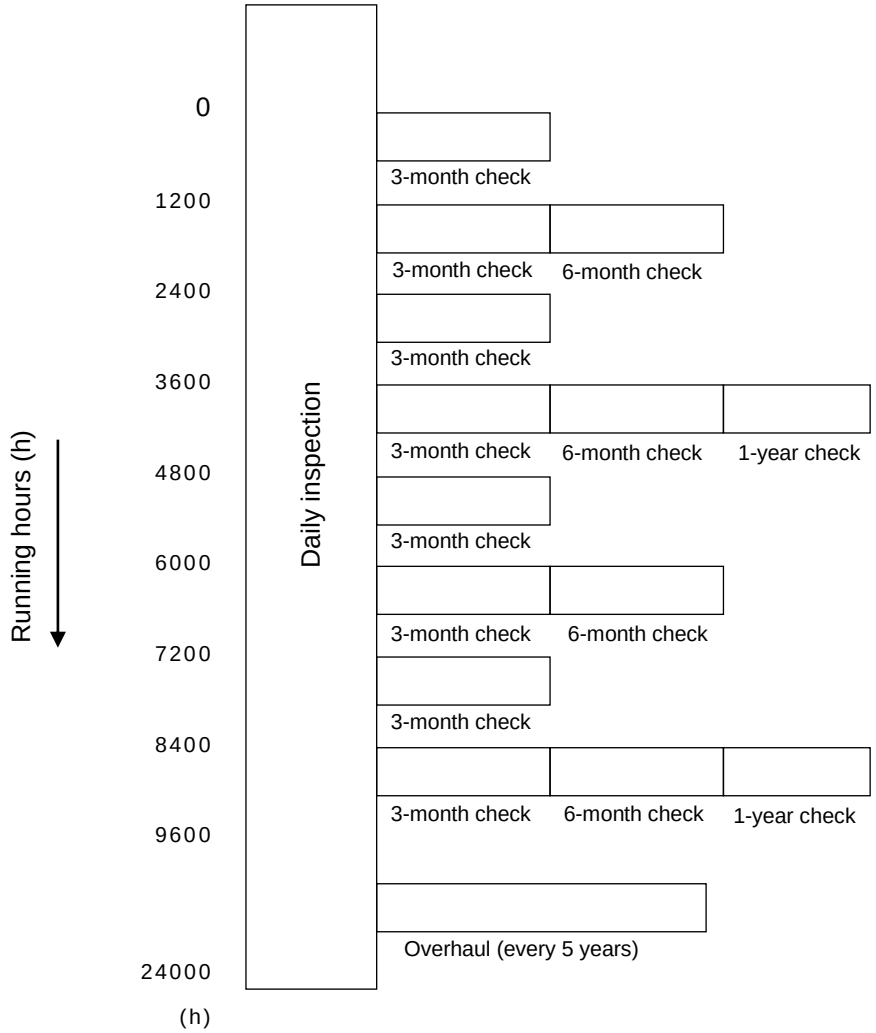
4. Robot Home Point and Position Detector Error 51
4.1 Setting Robot Home Point 51
4.2 Position Detector Error 52
4.3 Procedure for Reproducing Home point Data at Battery Restoration 53
4.3.1 Reproducing the home point data of axes 1, 2, 4 (rotation) and 5 (traverse) 53
4.3.2 Reproducing the zero-point data of axis 3 (Z) 55

5. Replacement Parts for Maintenance 59
5.1 Cautions for maintenance replacement parts 59
5.2 List of Replacement Parts – Main Robot (SR-404HC) 60
5.3 List of Replacement Parts – Main Robot (SR-404HC-Z150) 62
5.4 List of Replacement Parts – Main Robot (SR-624HC) 64
5.5 List of Replacement Parts – Main Robot (SR-405HC-L1550-R/L) 66
5.6 List of Replacement Parts – Main Robot (SR-405HC-L1530S-R/L) 68
5.7 List of Replacement Parts – Controller (TS3000, TS3000E) 70
5.8 List of Replacement Parts – Controller (TS3100, TS3100E) 71

1. Maintenance

1.1 Maintenance Schedule

Please do maintenance by a qualified person according to laws and regulations of each country. Maintenance comes in the two (2) types; daily inspection, and regular inspection and maintenance. For the regular inspection and maintenance, inspection items are added every 1,200 running hours.



Inspection schedule

- When the system is operated for 16 hours per day:
  1,200hours×16 hours x 25 days = 3 months
- When the system is operated for 24 hours per day:
  1,200hours×24 hours x 25 days = 2 months

If the system is operated for 16 hours per day, it should be inspected every three (3) months. If the number of hours of system operation per day becomes longer, the regular inspection on the system must be done at shorter intervals.

Fig.1.1 Maintenance Schedule

1.2 Items for Maintenance and Inspection

This section describes the items for maintenance and inspection.
For the executing procedures, see the relevant paragraph listed in the table below.

1.2.1 Inspection at Power OFF (at Non-Operation)

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection

Table 1.1 Inspection at Power OFF

Description	Check position	D	Q	S	A	Refer to
Make sure that all bolts are completely tightened and secured. Otherwise, tighten the bolts completely.	Tool set bolts	○	○	○	○	Para. 2.2.1
	Robot installation bolts	○	○	○	○	
Make sure that the connectors are locked.	Connector	○	○	○	○	Para. 2.2.2
Make sure that no cut or scratch is found. Clean the contaminant, if any.	Entire robot	○	○	○	○	
Make sure that the cables and air tubing are not worn out.	Cables and air tubing of each unit			○	○	Para. 2.2.5
Make sure that each timing belt is not worn off or cracked. Check each timing belt for tension.	Base interior			●	●	Para. 2.2.6
	Arm 1 interior			●	●	
	Arm 2 interior			●	●	
Make sure that the resin belt is not worn off or cracked. Check the resin belt for tension.	Traverse axis			○	○	Para. 2.2.7
Make sure that each axis can be operated properly by moving the axis by hand.	Entire robot			○	○	

*1: The items marked "●" may include some technical work.
It is difficult for the customer to implement such work, so please ask SHIBAURA MACHINE Service Department.

1.2.2 Inspection at Power ON (at Operation)

D: Daily inspection Q: Quarterly inspection
S: Semi-annual inspection A: Annual inspection
B: Biennial inspection

Table 1.2 Inspection at Power ON

Description	Check position	D	Q	S	A	B	Refer to
Make sure that no abnormal vibration or noise is caused.	Entire robot	○	○	○	○	○	Para. 2.2.3
Make sure that enough volume of grease is filled. Otherwise, fill the grease.	Axis 1 reduction gear				●	●	Para. 2.2.4
	Axis 3 ball screw		○	○	○	○	
	Axis 3 linear guide		○	○	○	○	
	Traverse axis ball screw *1		○	○	○	○	
	Traverse axis linear guide *1		○	○	○	○	
Push each arm by hand in the servo ON condition to make sure that the arm is secured.	Each joint				○	○	
Check for teaching position	Entire robot	○	○	○	○	○	
Replace the batteries for position detection with new ones.	Back side of the base					○	Para. 2.3

*1: Only for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L.

*2: The items marked "●" may include some technical work.
It is difficult for the customer to implement such work, so please ask SHIBAURA MACHINE Service Department.

1.2.3 Overhaul

Table 1.3 Overhaul

<Robot>

Description	Check position	Every 5 years	Refer to
Check for the consumable parts and replace as necessary.	Motors, belts, ball screws, reduction gears ,Harness	●	

< Controller >

Description	Check position	Every 5 years	Refer to
Replace the battery with a new one.	Backup battery in the controller	○	Para. 3.4.3
Replace the switching power supply unit with a new one.	In the controller	○	Para. 3.4.4

*1: The items marked "●" may include some technical work.
It is difficult for the customer to implement such work, so please ask SHIBAURA MACHINE Service Department.

1.3 Cautions on Maintenance and Inspection

Please do maintenance by a qualified person according to laws and regulations of each country. When performing inspection or maintenance of the robot, strictly observe the following precautions to protect yourself and coworkers.

DANGER

- Be sure to remove the power plug of the controller from the power source before approaching the robot for maintenance and inspection.
- When moving the robot by hand while the power is connected, be sure to assure the safe work and effect an emergency stop beforehand.

1.4 Maintenance Tools

We recommend using the following tools for maintaining the robot and controller.

- Screwdrivers (Phillips head screwdrivers, flat head screwdrivers)
- Hexagonal wrench keys, nominal size 2.5 mm to 14 mm
- Torque wrenches, nominal size 3 mm to 12 mm
- Torque driver • Wrenches, nominal size 5.5 mm, 7 mm, 46 mm, 50 mm
- Box wrench, nominal size 5.5 mm • Nippers • Needle-nose pliers • Pliers
- Torque wrench extension bar • Plastic hammer • Scale

Recommended provisions other than the tools:

- Alcohol (for cleaning) • Waste cloth • Spatula (for grease application)
- Loctite adhesive (242: mid to heavy-duty adhesive force) • Grease guns
- Liquid gasket(THREEBOND1221H) • Belt tension meter
- TSPC program creation/teaching support software (option)

1.5 Clamping Hexagon Socket Head Cap Screws and Set screws

This robot uses hexagon socket head cap screws at places requiring mechanical strength. At the time of factory-assembly, Loctite is applied and each screw is tightened with the following clamping torque.

When these screws and set screws are tightened again according to the inspection and maintenance as stipulated in this manual, use a torque wrench or torque driver, and Loctite (medium strength) to assure the appropriate clamping torque.

The places which do not fall under the table below are detailed in the paragraph of the replacement procedures.

Table 1.4 Clamping Torque

Hexagon socket head cap screw	–	–	M3	M4	M5	M6	M8	M10	M12
Set screws	M3	M4	M5	M6	M8	M10	M12	M16	M20
Clamping torque	1.5N·m	1.8N·m	2.0 N·m	4.7 N·m	9.0 N·m	15 N·m	37 N·m	75 N·m	128 N·m

For the screws arranged on a circle for mounting the reduction gear, etc., tighten them in the diagonal order, as shown below.

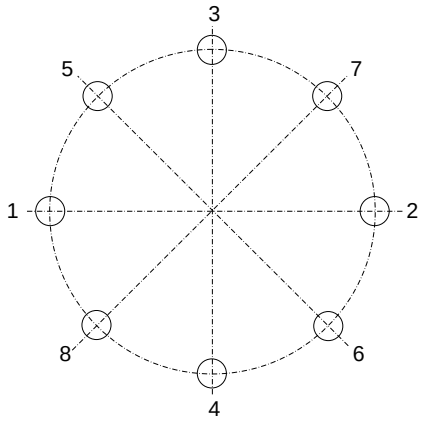


Fig.1.2 Clamping bolts on a circle

DO NOT tighten one (1) screw at a time. Tighten each screw in multiple steps, using a hexagonal wrench key, and secure with appropriate clamping torque by means of a torque wrench.

1.6 Maintenance Contract and Repair

1.6.1 Maintenance Contract

For the inspection items for maintenance and inspection to be done semi-annually and after, it is rather difficult for the user to execute, except for greasing and lubrication. We recommend that the user contact SHIBAURA MACHINE Service Department and conclude an after-sale service contract with us.

1.6.2 Repair

If a fault has occurred or is time, advise us of the details of the fault and the following information stated on the robot and controller.
(For the contact information of the SHIBAURA MACHINE Service Department, please refer to Service Network.)

- Robot model
- Manufacture number
- Manufacture number of controller
- Date of manufacture of robot
- Running hours

For the manufacture number and date of manufacture, see the following labels attached to main base of the robot.

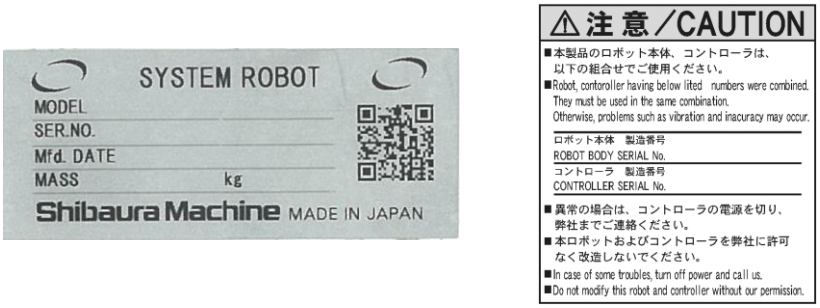


Fig.1.3 Label

1.6.3 Modification

The robot and controller MUST NOT be modified or disassembled without a prior consent from SHIBAURA MACHINE.

! CAUTION

- The user must NEVER replace or modify parts other than those described in the instruction manual. Otherwise, the performance may deteriorate or faults or accidents will be caused.

2. Maintenance of the Main Robot

2.1 Layout of Robot Components and Drive Mechanism

The layout of the robot mechanical components is shown in Fig. 2.1.

a) Axis 1

The output of axis 1 AC servo motor is input to the RV reduction gear (made by Nabtesco) via the spur gear. The output of the RV reduction gear directly controls the arm 1 motion.

b) Axis 2

The output of axis 2 AC servo motor is input to the harmonic drive via the timing belt. The output of the harmonic drive directly controls the arm 2 motion.

c) Axis 3

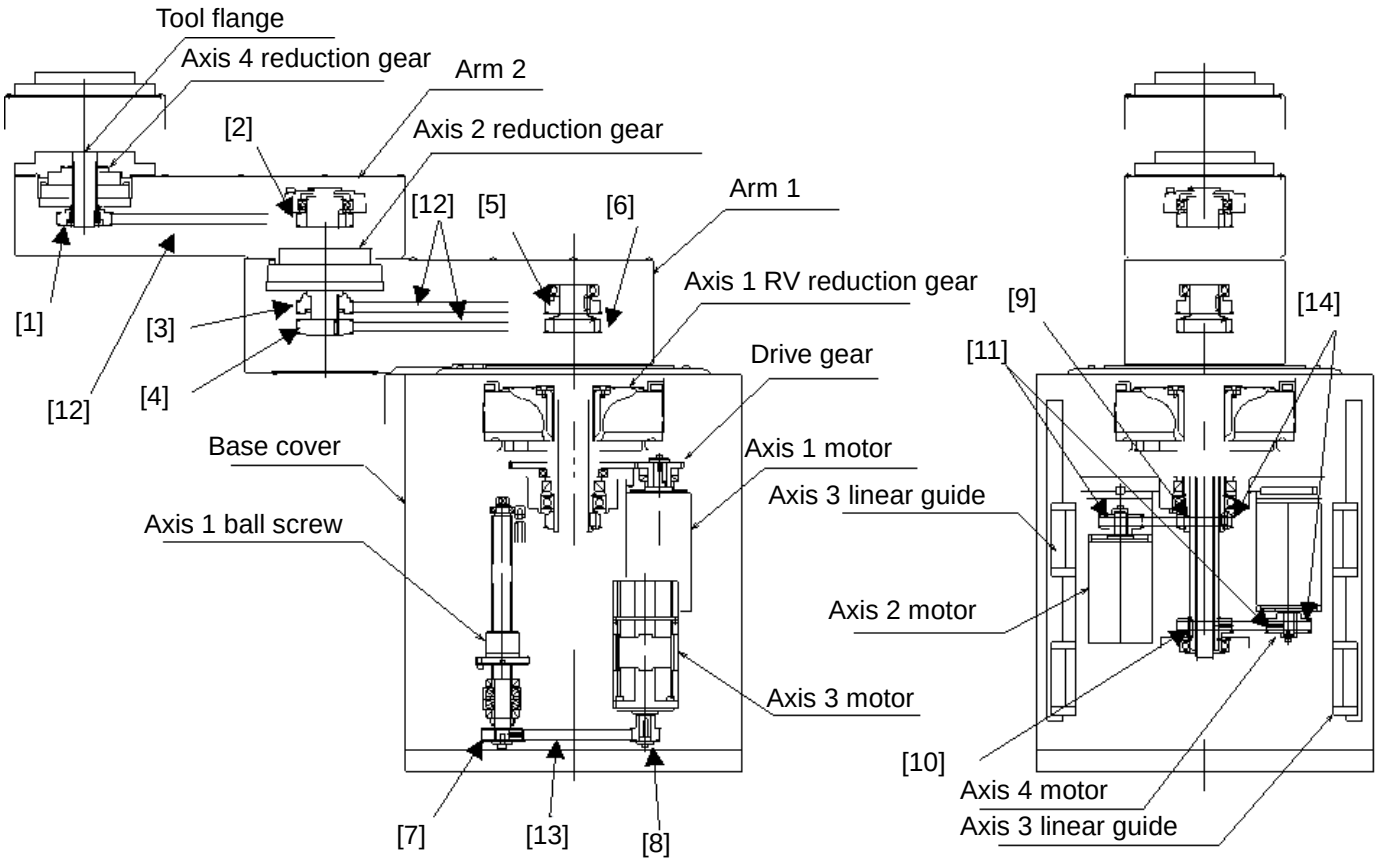
The output of axis 3 AC servo motor is input to rotate the ball screw via the timing belt to move the vertical axis up and down. The axis 3 simultaneously moves up and down the three (3) motions of axis 1 arm, axis 2 arm and axis 4 hand.

d) Axis 4

The output of axis 4 AC servo motor is input to the harmonic drive via the timing belt. The output of the harmonic drive directly controls the axis 4 hand rotation.

e) Axis 5 (only for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

The output of axis 5 AC servo motor is input to rotate the ball screw. The output of the ball screw directly controls the slider motion.



Part numbers [1] to [11] are the timing pulleys and part numbers [12] to [14] the timing belts.

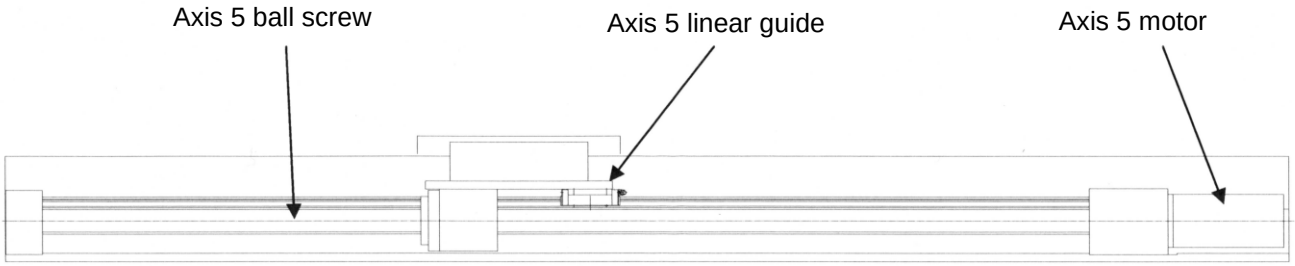


Fig.2.1 Layout of robot mechanical components

2.2 Details of Inspection

2.2.1 Check of Each Bolt (or Screw) for Clamping

1) Tool (hand) Set Bolts

Using the hexagonal wrench key, make sure that the clamping bolts of the tool set flange, which are clamped to the arm 2, are tightened completely. If loose, tighten them completely.

Using the hexagonal wrench key, make sure that the mounting bolts, which secure the tool (hand) to the tool flange, are tightened completely. If loose, tighten them completely.

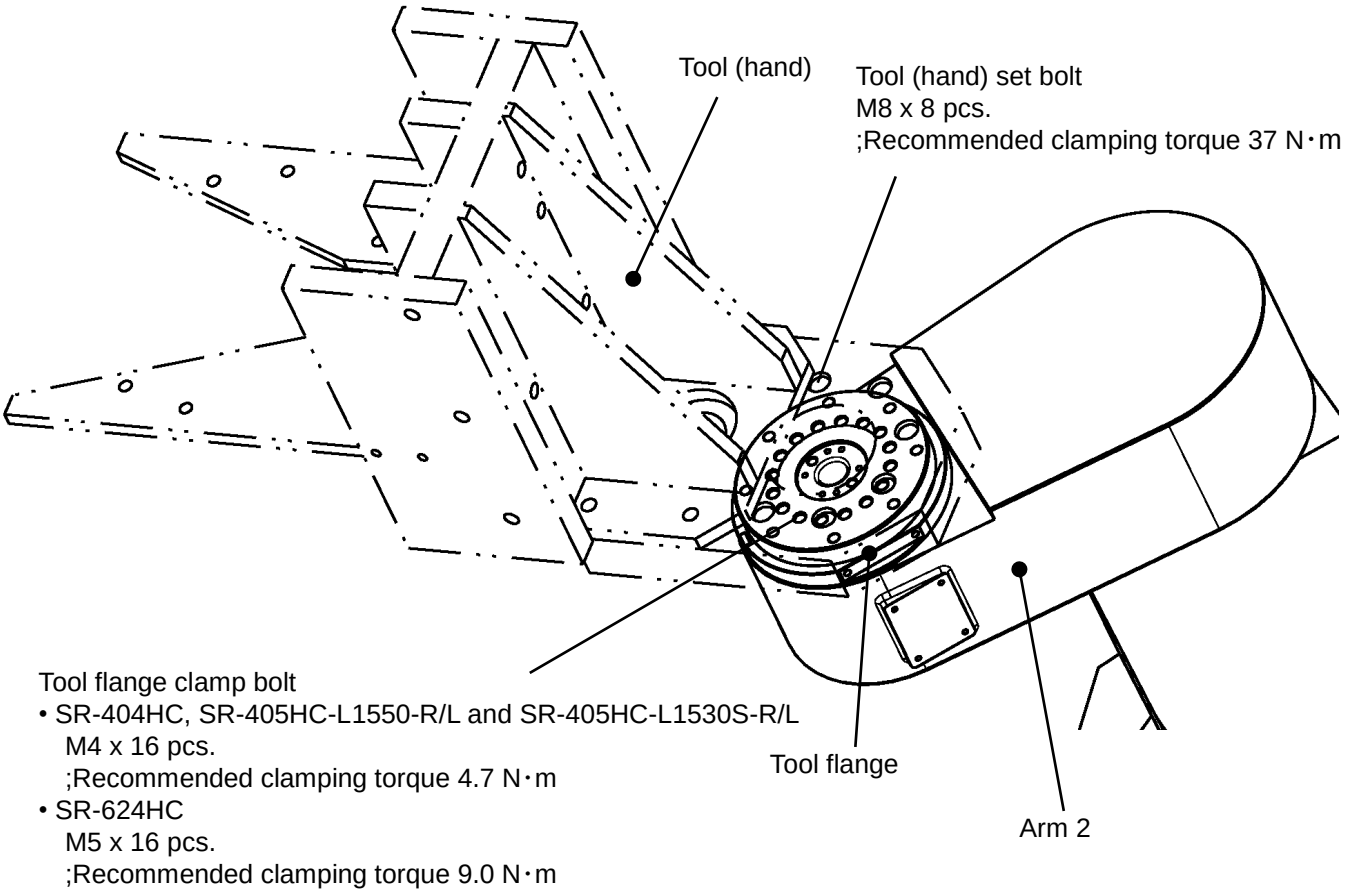


Fig. 2.2 Tool (hand) set bolts

2) Robot Installation Bolts

Make sure, using the hexagonal wrench key, that the installation bolts of the main robot base are tightened completely. If loosened, tighten them completely.

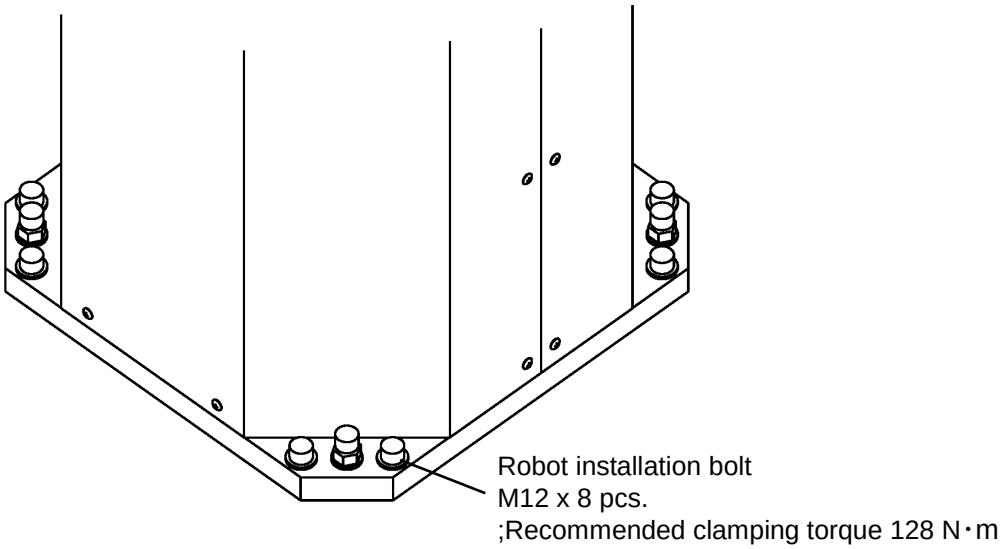


Fig. 2.3 Robot installation bolts (SR-404HC and SR-624HC)

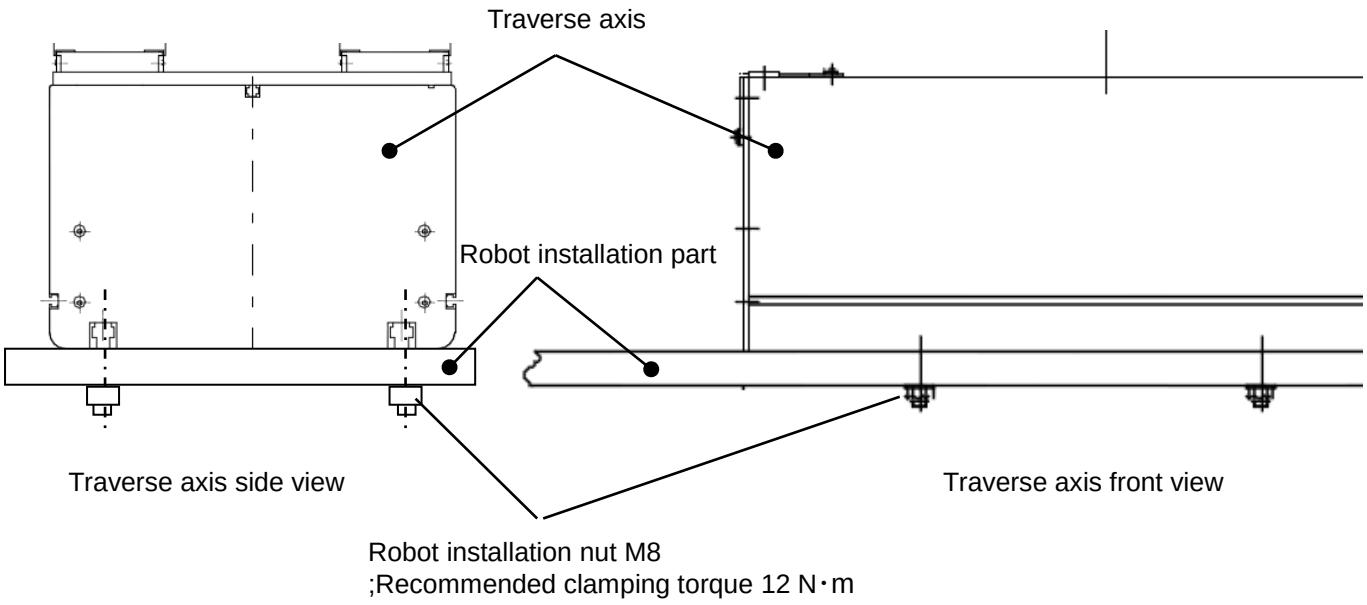


Fig. 2.4 Robot installation bolts (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

2.2.2 Check of Each Connector for Clamping

Make sure that the cable connectors connected to the robot main base and traverse axis are tightened completely. If loosened, tighten them completely.

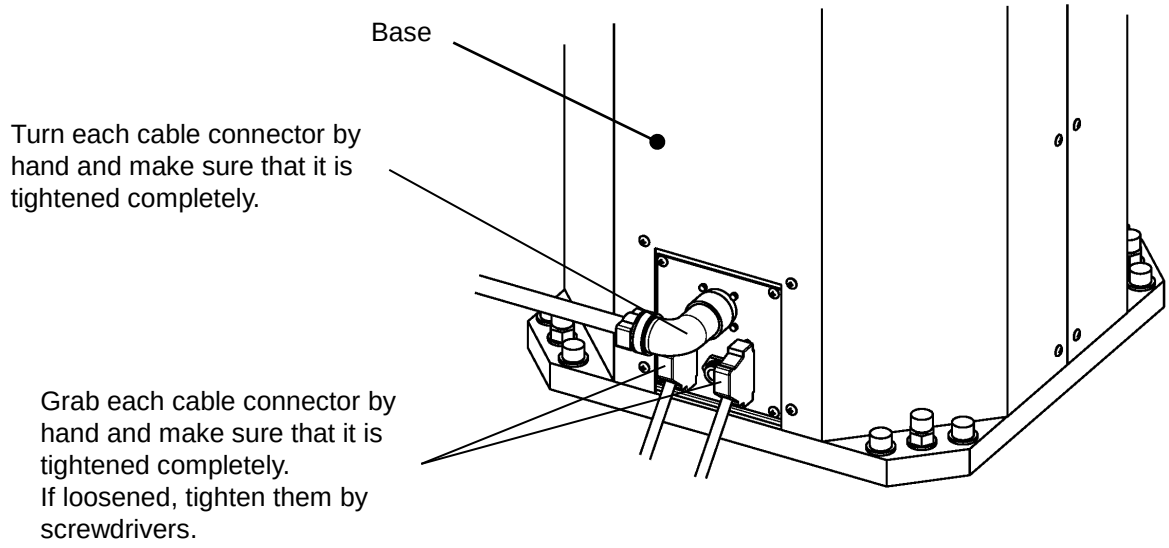


Fig. 2.5 Base connectors (SR-404HC and SR-624HC)

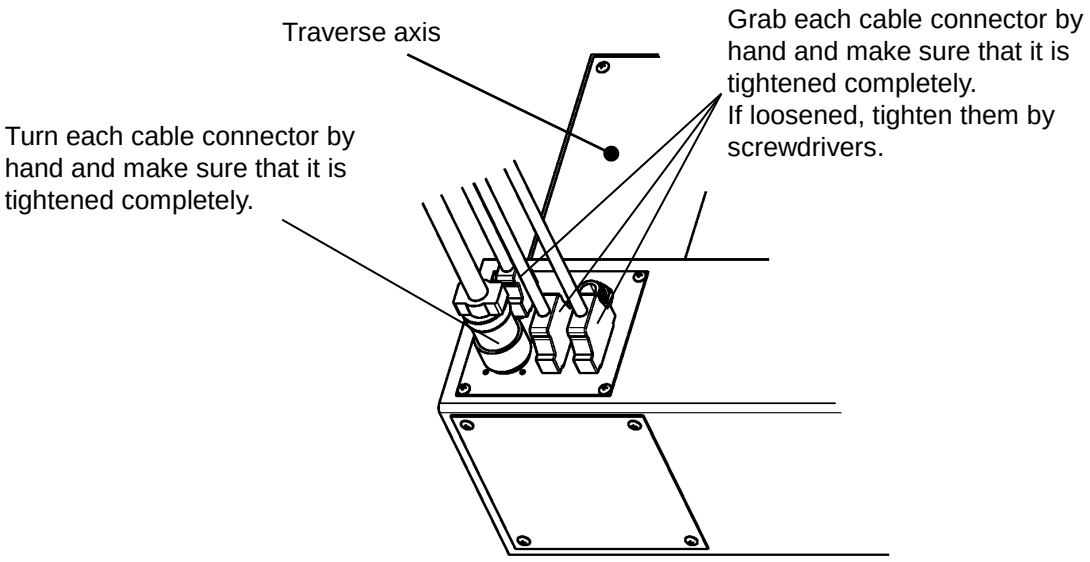


Fig. 2.6 Traverse axis connectors (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

2.2.3 Check for Vibration and Abnormal Noise

1) Approach the robot and its peripheral equipment outside the predetermined operating range and make sure that no abnormal noise of operation is heard. Otherwise, locate the source of the abnormal noise.
When necessary, turn off the power of the robot and its peripheral equipment and manually move the part suspected to be the trouble source to confirm that the abnormal noise is heard from it.
If the abnormal noise is heard from the ball screw and linear guide, contaminant has possibly entered them, or they have run out of grease. For the former cause, remove the contaminant and make sure that other parts are not damaged. For the latter cause, apply grease.

2) If chattering is caused in the cover, etc., make sure that all screws are tightened completely and that relevant parts are not deformed. Otherwise, completely tighten the screws or repair the parts as necessary.

3) Make sure that the screws of each arm joint are tightened completely. Otherwise, tighten the screws completely.

4) Possible causes other than the above-mentioned are the wrong method of use, controller-related cause, etc. In either case, adjustment is not easy. Consult us at the after-sale service agent in your territory.



DANGER

- When nearing the robot in the servo-free status, be sure to carry out the work by two (2) persons. One person should carry out the work while the other is monitoring the work outside the hazardous zone. The latter person should watch the work and be ready to press the emergency stop switch if any abnormality occurs.

2.2.4 Check for Lubrication

For the RV reduction gear of axis 1, use the Moly-White grease RE No.00 (made by Nabtesco).
For the axis 3 (vertical axis), use AFE grease (made by THK).

For the reduction gear in the arm, the following grease is used. The reduction gear need not be greased again, however. If an abnormal noise or backlash is found before overhaul, consult us at the after-sale service agent in your territory.

Grease for arm reduction gear: 4B No.2 (Harmonic drive)

1) Check of ball screw and linear guide for lubrication state
Make sure that the ball screw and linear guide of each axis are greased sufficiently and that no rust has developed on them. If they have run out of grease, fill it.

! CAUTION

- NEVER mistake the kind of grease. Otherwise, the robot performance will deteriorate or the robot will malfunction.
- After you touched the ball screw, ball screw spline or linear guide by hand, be sure to fill the grease. Otherwise, rust will develop on it.

2) Greasing

i) Dismounting of base cover:

When you remove the two-split base cover, you can see the linear guide there.

Move axis 3 (Z) of the robot to the upper limit position and remove the base cover.

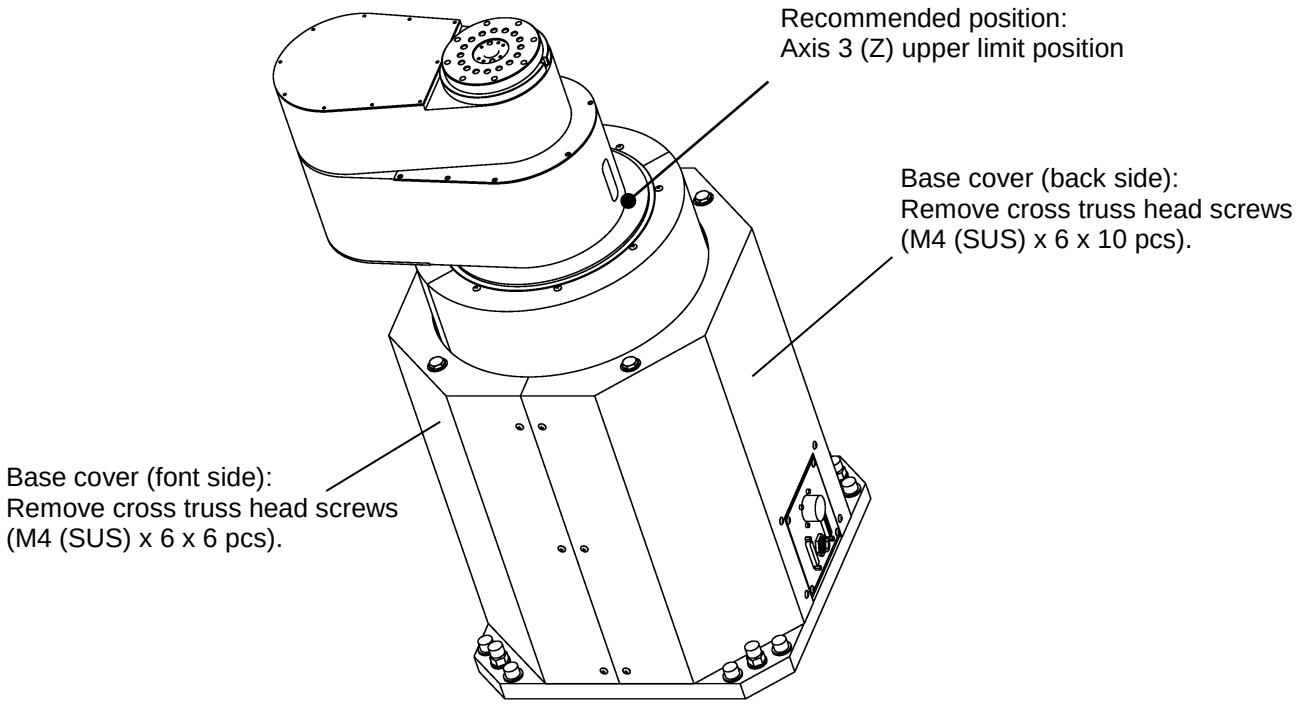


Fig. 2.7 Dismounting of base cover

ii) Greasing of linear guide of the axis 3(Z):

Apply the grease to the rails.

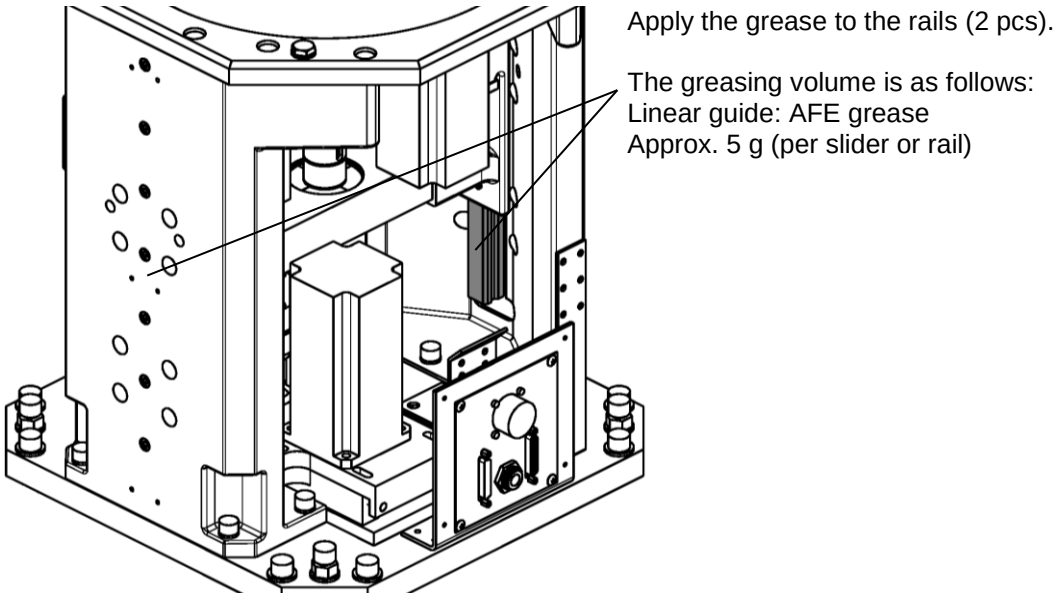


Fig. 2.8 Greasing of linear guide

iii) Greasing of ball screw of the axis 3 (Z):

As shown in Fig. 2.9, move the axis 3 (Z) to a position where the grease nipple of the ball screw nut can be seen through the grease supply hole, and supply grease through the grease nipple by grease guns.

The greasing volume is as follows:

Ball screw: AFE grease Approx. 5 g

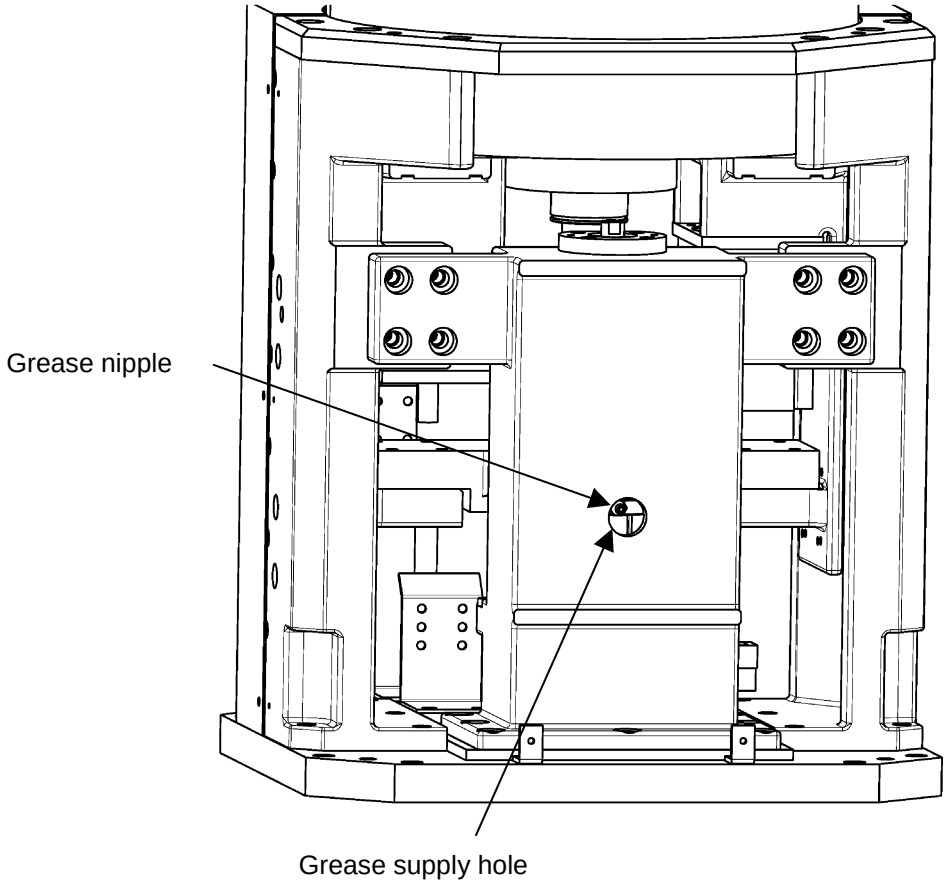


Fig. 2.9 Greasing the ball screw

iv) Mounting of base cover:

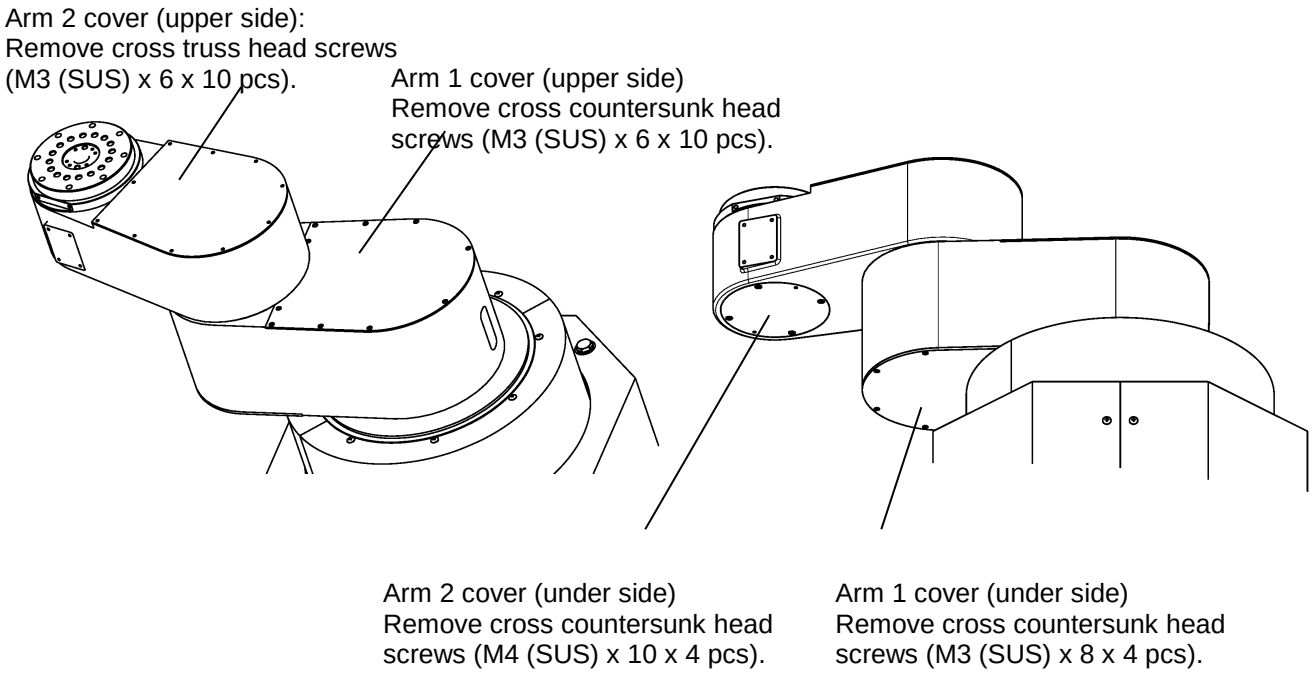
Set the two-split base cover.

v) Greasing of ball screw and linear guides of traverse axis (Only for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L):

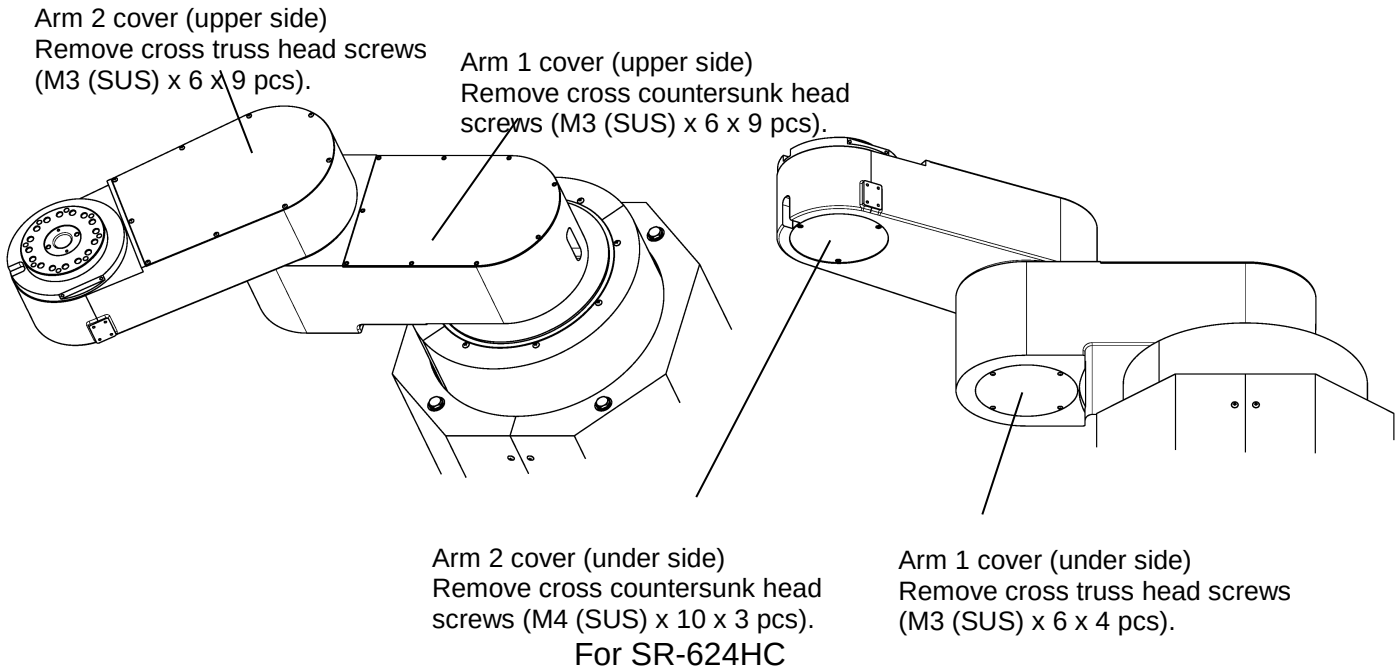
See the ROIBOT actuator instruction manual provided separately.

2.2.5 Check for Wiring

Make sure that the cable surface in the base cover is not damaged and in a good state. Open the arm 1 and arm 2 covers and make sure that each cable surface is not damaged, or stretched or slackened due to unusual force exerted on it.



For SR-404HC, SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L



For SR-624HC

Fig. 2.10 Removing the arm 1 and arm 2 cover

2.2.6 Check for Timing Belt Status

Remove the cover and visually check to see that the belt is not worn off or cracked. Also check the timing belt for tension and adjust as necessary.

2.2.7 Check for Resin Belt
(Only for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

Make sure that the resin belt of the traverse axis is not loosened or damaged.
To replace the belt, see the ROIBOT actuator instruction manual provided separately.

2.3 Replacing Position Detector (or Encoder) Batteries

CAUTION

- The batteries should be disposed of according to the user's in-house regulations. NEVER drop the battery into fire. NEVER short-circuit, charge, disassemble or heat it. Otherwise, liquid leakage or rupture may be caused.

To keep the data of the position detector attached to the motor, they are backed up by batteries. Be sure to replace the batteries with new ones at the time of biennial inspection. When the robot is not used for a long term (i.e., the batteries are left intact in the power OFF condition), replace the batteries at the time of robot startup. As a yardstick, the shutdown period is two (2) months.

Note: Dedicated batteries must be used. For battery replacement, see " Section 5, Replacement Parts for Maintenance" of this manual and contact SHIBAURA MACHINE.

CAUTION

- When the battery voltage has dropped, a battery alarm will generate. If the batteries are replaced just after generation of the battery alarm, the battery voltage returns to normal with the battery alarm reset automatically. Unless the batteries are changed just after generation of the battery alarm, however, the battery voltage will drop further and a battery error will occur. Under this condition, position data detected by the encoder is not reliable. As a result, a position detection error occurs and the robot enters an emergency stop state so that the servo system cannot be turned on. If the power is turned off in this condition, the position data is lost.

  To avoid the above, be sure to replace the batteries with new ones at the time of biennial inspection.

2.3.1 Replacing Batteries

DANGER

- Only after assuring the safe work, replace the batteries with new ones while the power is turned on and an emergency stop is effected on the robot.

1) Assure the safe work. Then keep the power ON and press the EMERGENCY pushbutton switch to effect an emergency stop on the robot.
2) Remove the base cover (back side) as shown in Fig. 2.11.
3) Replace the batteries within five minutes. If replacing them when the power is off, let at least one minute pass after turning off the power. (This is because current flows for several tens of seconds after power-off.) Then, start and complete replacement within five minutes.
4) Load the new batteries, and mount the base cover (back side).

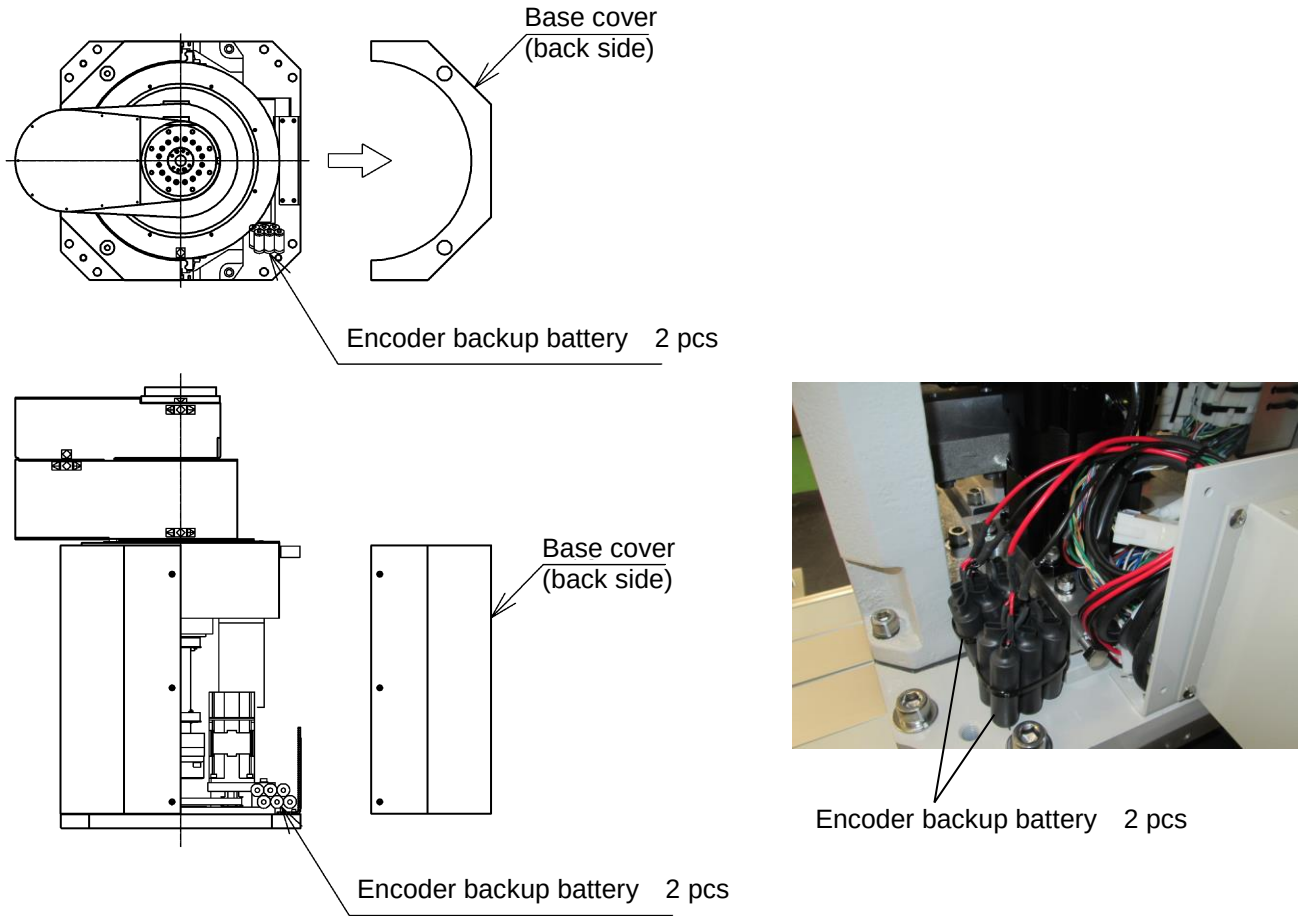


Fig. 2.11 Replacing the batteries for the position detector

2.3.2 Battery Error Code

When a position detector error including battery error has occurred, the following error code is shown on the error display.

When electric voltage decreased, the following error code is shown on the error display. When the following error code is shown in the error history at generation of a position detector error, the error can be identified as the battery error.

Table 2.1 1-40 x battery error code

Error code	Error content	Remarks
1-401	Axis1 Enc Battery low (Battery alarm)	
1-402	Axis2 Enc Battery low (Battery alarm)	
1-403	Axis3 Enc Battery low (Battery alarm)	
1-404	Axis4 Enc Battery low (Battery alarm)	
1-405	Axis5 Enc Battery low (Battery alarm)	Case of TS3100 or TS3100E (for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

If the error in table 2.1 occurred, please replace battery immediately. If unattended, level 8 errors such as listed in Table 2.2 to 2.4 occur. In addition, it is possible to lost robot origin position information.

Table 2.2 8-60 x battery error code

Error code	Error content	Remarks
8-601	Axis1 Enc multi count err	
8-602	Axis2 Enc multi count err	
8-603	Axis3 Enc multi count err	
8-604	Axis4 Enc multi count err	
8-605	Axis5 Enc multi count err	Case of TS3100 or TS3100E (for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

If the error in table 2.2 displayed, counter skipping may be occurring due to insufficient battery power. Change the battery. If changing the battery does not correct the problem and the error is still displayed, the encoder may be faulty. In such a case, it is necessary to change the motor. Please consult our service representative.

Table 2.3 8-61 x battery error code

Error code	Error content	Remarks
8-611	Axis1 Enc battery empty	
8-612	Axis2 Enc battery empty	
8-613	Axis3 Enc battery empty	
8-614	Axis4 Enc battery empty	
8-615	Axis5 Enc battery empty	Case of TS3100 or TS3100E (for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

If this error is displayed, the battery power may be insufficient. Change the battery. And it is possible to lose robot origin position information because the battery power was insufficient. If alarm does not clear after you replace battery, Please set origin position by referring to “Section 4, Robot Home Point and Position Detector Error.”

Table 2.4 8-40 x battery error code

Error code	Error content	Remarks
8-401	Axis1 Encoder abnormal	
8-402	Axis2 Encoder abnormal	
8-403	Axis3 Encoder abnormal	
8-404	Axis4 Encoder abnormal	
8-405	Axis5 Encoder abnormal	Case of TS3100 or TS3100E (for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

In the case of level 8 battery errors, the above two types errors in table 2.2 and 2.3 are basically displayed; however, the errors in table 2.4 may be displayed in rare cases. If this error is displayed, change the battery. And it is possible to lose robot origin position information because the battery power was insufficient. If alarm does not clear after you replace battery, Please set origin position by referring to “Section 4, Robot Home Point and Position Detector Error.”.

The battery error is one of the position detector errors (i.e., encoder errors), and there is other cause leading to the position detector error. The restoring method from the position detector error varies with the error contents generated. To properly execute the restoring method from the position detector error, therefore, read through and completely understand the descriptions carried in “Section 4, Robot Home Point and Position Detector Error.”

3 Maintenance of Controller

3.1 Cautions on Maintenance and Inspection

When performing maintenance and inspection of the controller, follow the items given below so that the work can be carried out safely.

! CAUTION

- Before removing the controller cover for maintenance or inspection, be sure to turn off the main power switch of the controller.
  Also, turn off the power breaker when connecting or disconnecting the power cable.
- DO NOT touch the servo driver for at least two (2) minutes after the main power switch is turned off. The large-capacity capacitor in the servo driver may be charged with voltage, and touching the servo driver could lead to electric shocks.
- DO NOT disconnect the battery connector at other than battery replacement. Otherwise, files saved in the memory may be lost.

! DANGER

- Be sure to turn off the main power switch. If the main power switch remains ON, the servo power printed board, servo printed board and switching power supply are charged, and touching them could lead to electric shocks.

3.2 Layout of Controller Parts (TS3000, TS3000E)

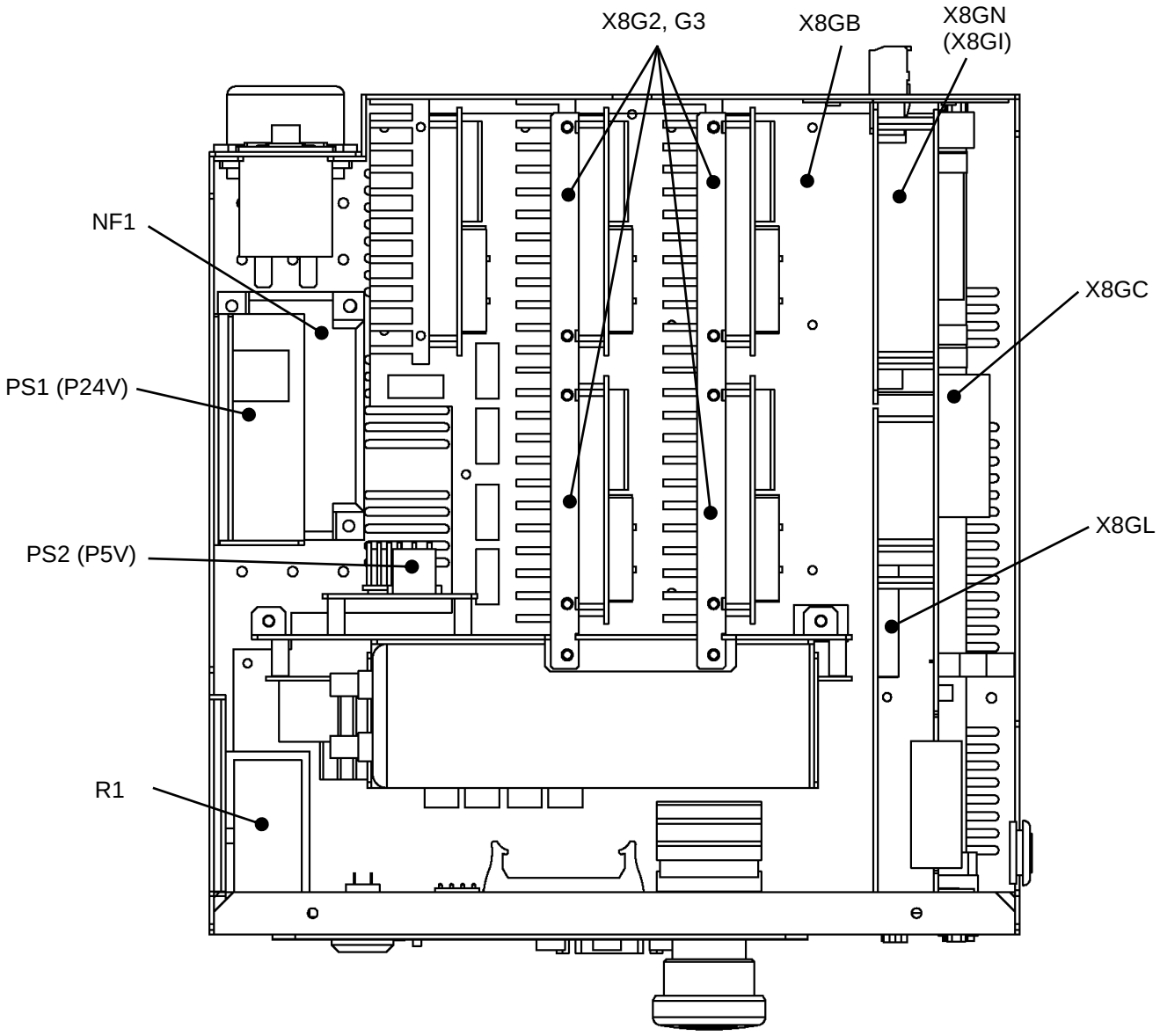


Fig. 3.1 Layout of controller parts (TS3000, TS3000E)

Table 3.1 Controller parts (TS3000, TS3000E)

Part name	Descriptions
PS1	P24V output switching power supply
PS2	P5V output switching power supply
X8GC	Main control printed board
X8GN (X8GI)	I/O output printed board
X8GL	Servo logic printed circuit board
X8G1~X8G3	Servo amplifier printed circuit board
X8GB	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.3 Layout of Controller Parts (TS3100, TS3100E)

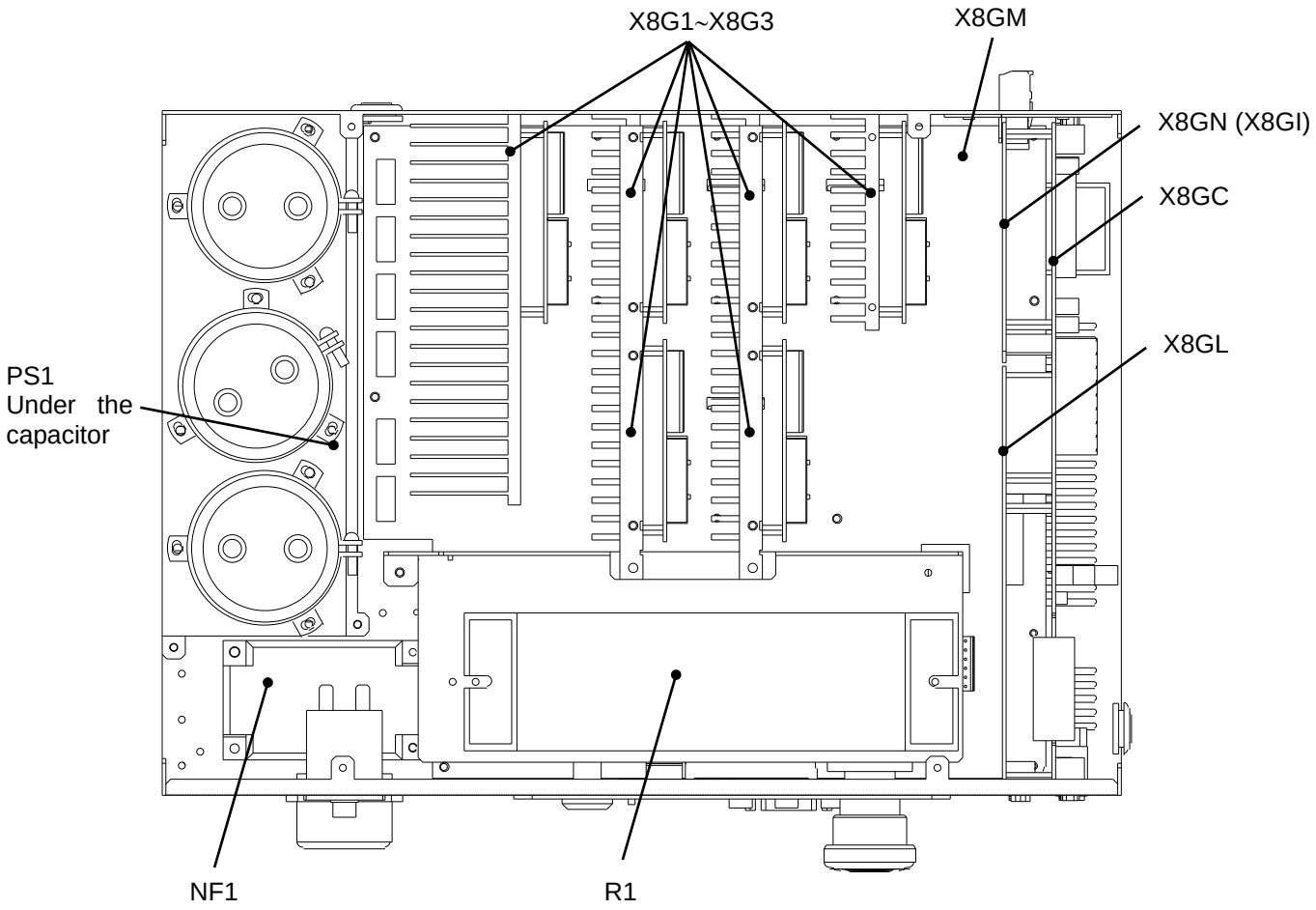


Fig. 3.2 Layout of controller parts (TS3100, TS3100E)

Table 3.2 Controller parts (TS3100, TS3100E)

Part name	Descriptions
PS1	P5V, P24V switching power supply
X8GC	Main control printed board
X8GN (X8GI)	I/O output printed board
X8GL	Servo printed board
X8G11~ X8G3	Servo amplifier board
X8GM	Mother board
R1	Regenerative resistor
NF1	Noise filter

3.4 Maintenance Procedures

3.4.1 Check for Controller Air Vent Holes

If the air vent holes are blocked, the controller may overheat and malfunction.
To avoid this, perform check on the air vent holes to make sure that air is flowing freely through them.

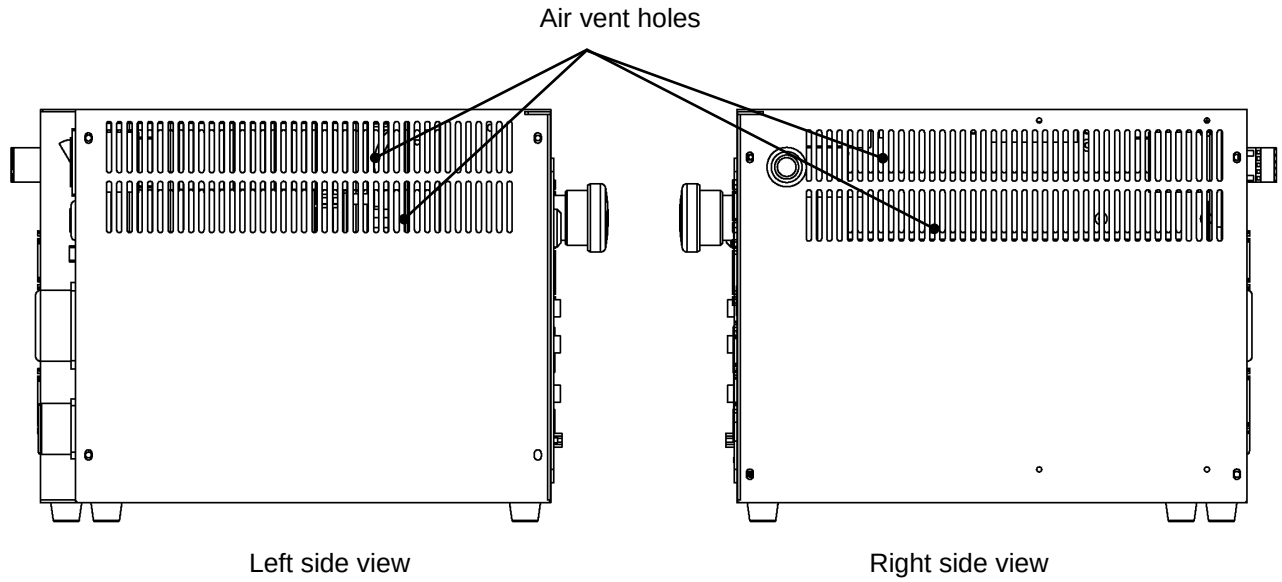


Fig. 3.3 Controller side views (TS3000, TS3000E)

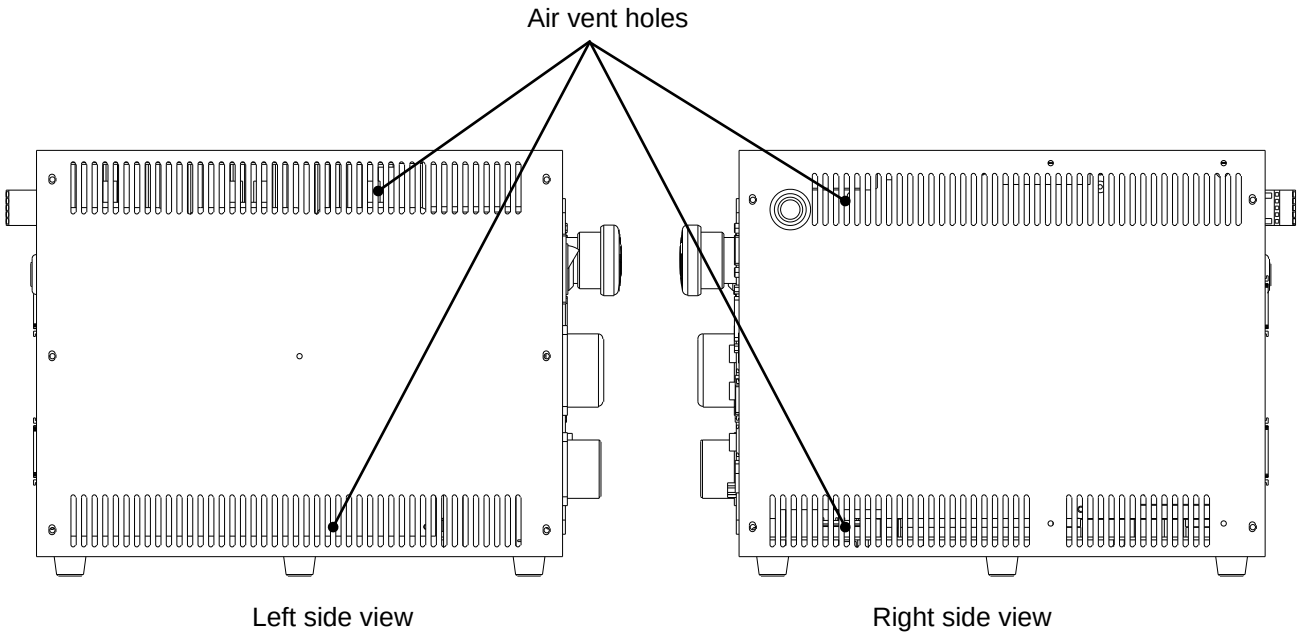


Fig. 3.4 Controller side views (TS3100, TS3100E)

a) If there is any obstacle, move them away from the vent holes so that the air flow is not blocked.

b) Make sure that no contaminant is left on the air vent holes. If any contaminant is let on the vent hole, remove it.

c) If the controller is placed sideways, make sure that the air vent holes on the bottom surface of the controller are more than 11 mm away from the floor surface. Always keep the space of 11 mm or over. The length of rubber shoe is approximately 11 mm.

3.4.2 Check of Safety Devices for Function

Make sure that the emergency stop pushbutton switches equipped on the control panel and teach pendant work properly. Also make sure that the safety devices controlled by the external operation input signals work correctly.

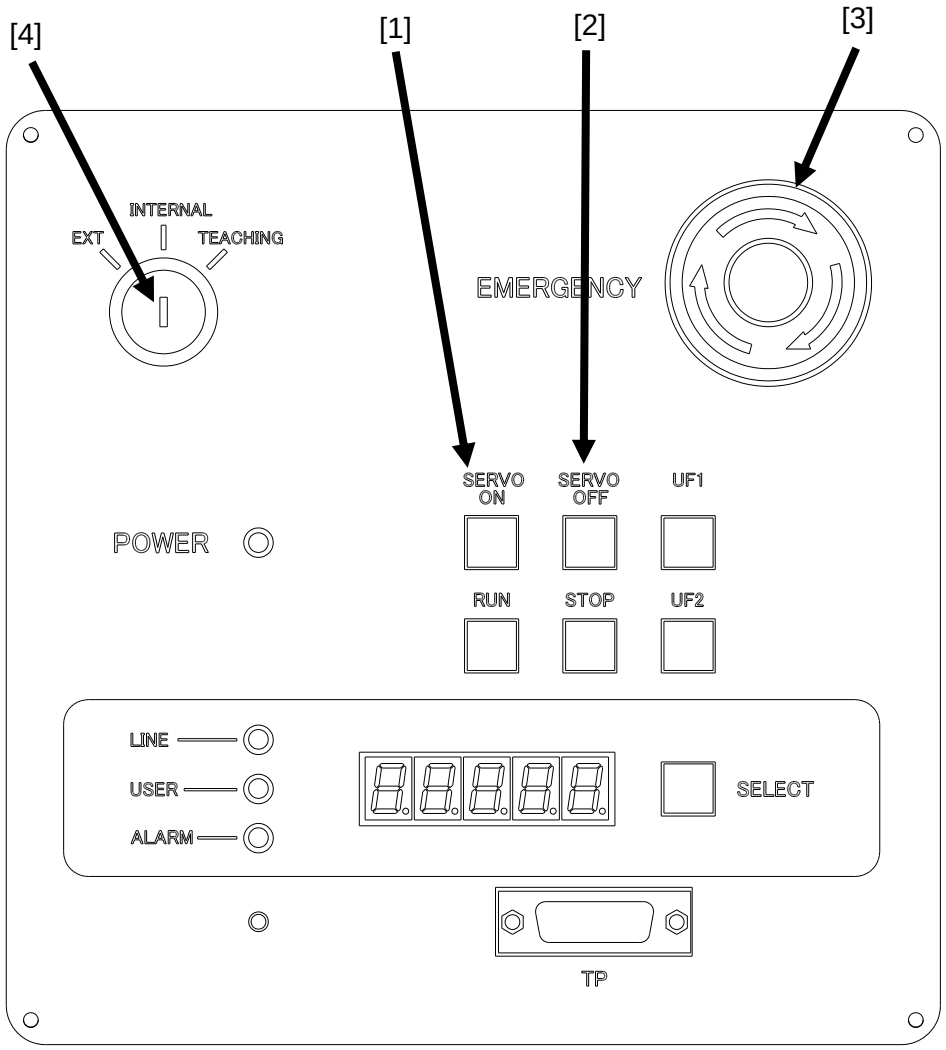


Fig. 3.5 Layout of control panel switches

a) Make sure that the EMERGENCY (emergency stop) switch [3] provided on the control panel functions properly.

1) Set the key switch [4] to the INTERNAL mode.

2) Press the SERVO ON switch [1] provided on the control panel to turn the servos on and make sure that the SERVO ON switch [1] LED is illuminated.

3) Press the EMERGENCY switch [3] and make sure that the servos are turned off. The SERVO ON switch [1] LED goes out and the SERVO OFF switch [2] LED illuminates. At this time, make sure that the EMERGENCY switch [3] remains depressed.

4) Turn the EMERGENCY switch [3] clockwise and make sure that the same switch turns off.

b) Make sure that the EMERGENCY switch [6] provided on the teach pendant functions properly.

1) Turn around the key at front of controller for selecting the TEACHING.

2) To turn on the servo power, press the SERVO ON switch [5] on the teach pendant while keeping the ENABLE switch at the neutral position. Make sure that the SERVO ON switch LED [5] is illuminated.

3) Press the EMERGENCY switch [6] on the teach pendant and make sure that the servos are turned off. The SERVO ON switch LED [5] goes out. At this time, make sure that the EMERGENCY switch [6] remains depressed.

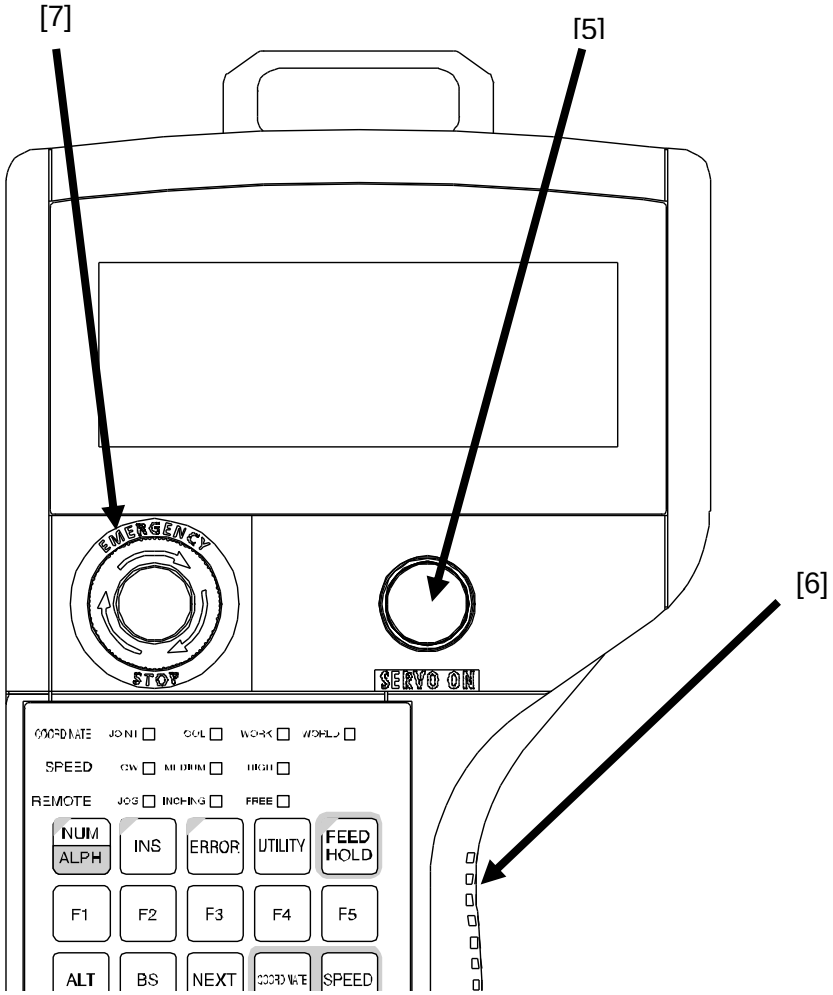


Fig. 3.6 EMERGENCY switch provided on teach pendant

4) Turn the EMERGENCY switch [7] in the clockwise (CW) direction and make sure that the same switch turns off.

c) Make sure that the safety devices controlled by the external operation signals work properly.

1) Press the SERVO ON switch [1] equipped on the control panel to turn on the servo power. Then turn on (or trip) the safety devices connected with external operation input signal "emergency stop" (such as external emergency stop switch, photoelectric type sensing safety device and foot switch) to make sure that the servo power is turned off.

2) Set ON the safety devices connected with external operation input signal "low speed command", and make sure that each signal functions properly.

3.4.3 Battery Replacement

The memory equipped on the X8GC printed board of the controller is backed up by a lithium battery to save the data. Replace the battery every five (5) years. The lithium battery will turn its life when used for a predetermined time. If it is used, neglecting the life, the battery voltage will drop to below the voltage required for keeping the memory data, resulting in data being lost and faults caused by leakage of the battery liquid.

If the following alarm has occurred, replace the battery with a new one.

"MAIN Battery alarm" (alarm code 1–145)

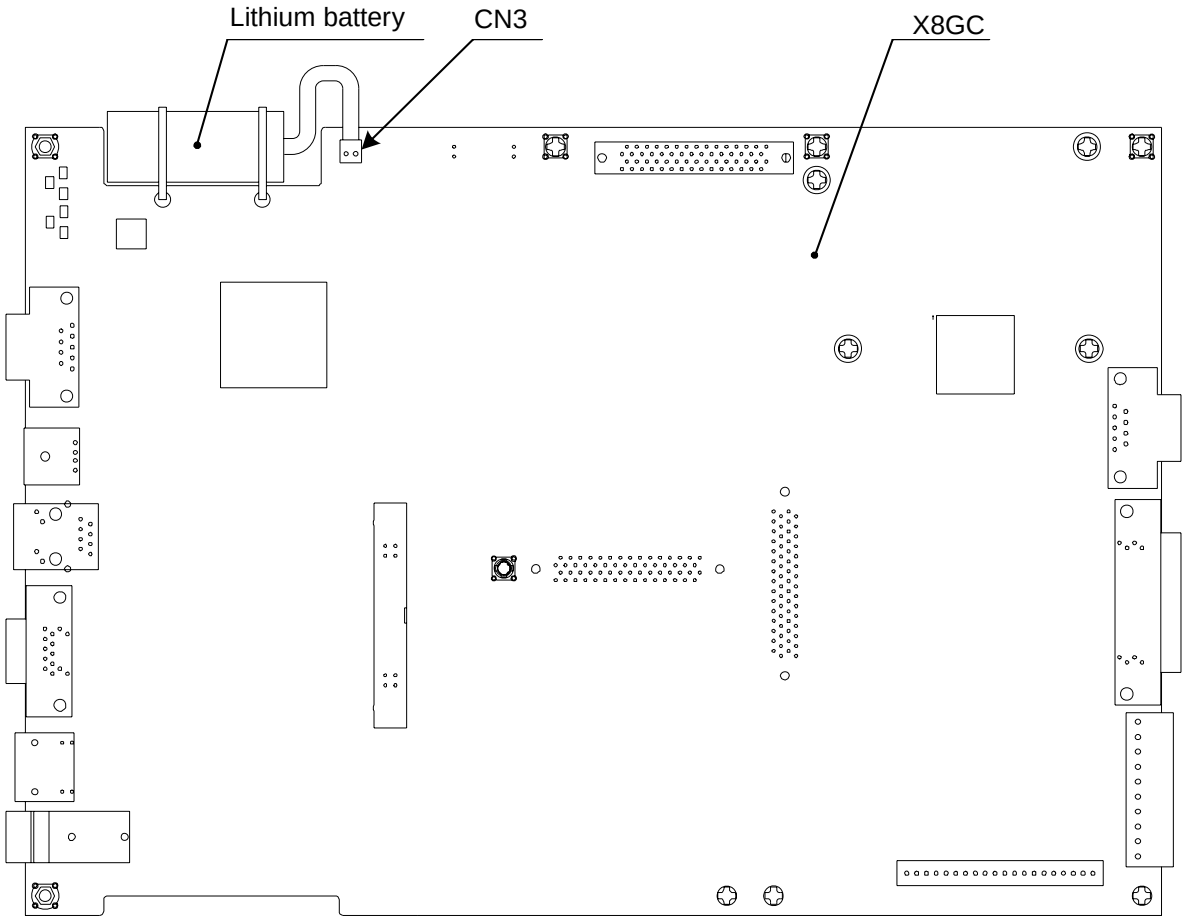


Fig. 3.7 Location of lithium battery

Table 3.3 Type of battery

Type of battery	Maker	Battery service life	Remarks
ER6C WK27	Hitachi Maxell	10 years (at room temperature)	Discontinued in 2022
ER6V C4	TOSHIBA	10 years (at room temperature)	Adopted starting in 2022

Note: The battery life is ten (10) years at the room temperature. As it differs with the external environment such as operating temperature and humidity, we recommend to replace the battery every five (5) years.
Only the battery shown in Table 3.3 should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

Battery replacement procedures

1) If the battery is kept removed for more than one (1) minute, save all programs or parameters stored in the memory into the personal computer. In the worst case, all memory data are lost.

2) Set OFF the MAIN POWER switch.

3) Have a new battery ready, remove cross truss head screws that secure the cover, and then remove the cover from the main body.

4) Remove the battery connector from CN3 of the X8GC printed board, then disconnect the battery from the Tie-wrap securing the battery.
(For the battery position, see Fig. 3.7.)

5) Set a new battery immediately.
(If the controller is left for one minute or more with the battery being disconnected, all the memory data is lost.)

The lithium battery capacity decreases by natural discharge even if not used. Use a lithium battery which is within three (3) years (kept at the room temperature) from purchasing.

6) Mount the cover to the main body using cross truss head screws.

7) If the battery is kept removed for more than one (1) minute, set ON the main power switch and load the programs from the personal computer that has TSPC (option) installed.
If the controller has detected the battery voltage drop alarm even before the five (5)-year replacement period, replace the battery immediately.

! CAUTION

- Batteries should be disposed of according to the user's in-house regulations.
- NEVER drop the batteries into fire. NEVER short-circuit, charge, disassemble or heat them. Otherwise, liquid leakage or rupture may be caused.

3.4.4 Replacement of Switching Power Supply Unit

The life of the switching power supply unit (DC5 V/DC24 V) used in the controller differs with the operating conditions. This unit uses an aluminum electrolytic capacitor, and if the load current is large, the running time is long and the ambient temperature is high, the life will reduce.

As the life varies largely with the user's operating conditions (5 ~ 10 years), replace the power supply unit every five (5) years for preventive maintenance.

[PS1 Replacement procedures]

1) Turn off the main power switch.
2) Turn off the power breaker.
3) Remove the cover from the controller.

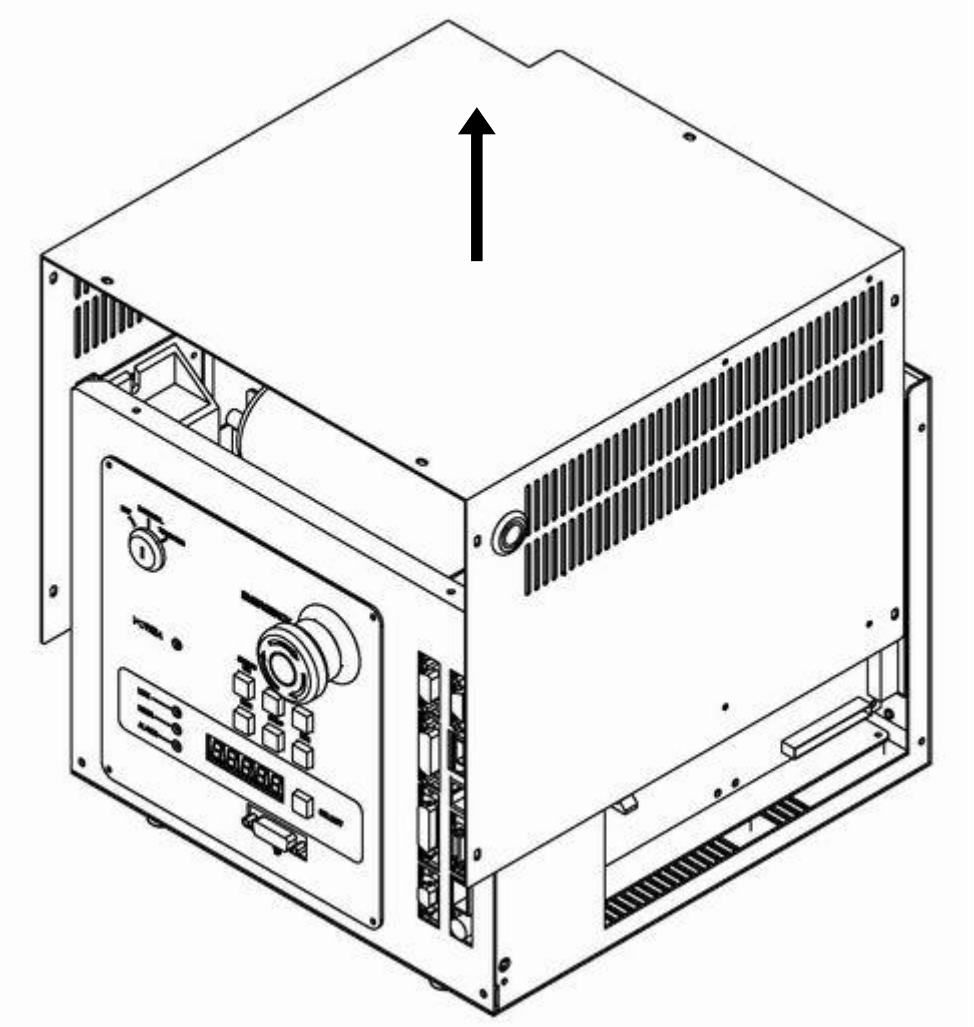


Fig. 3.8 Removal of upper cover (TS3000, TS3000E)

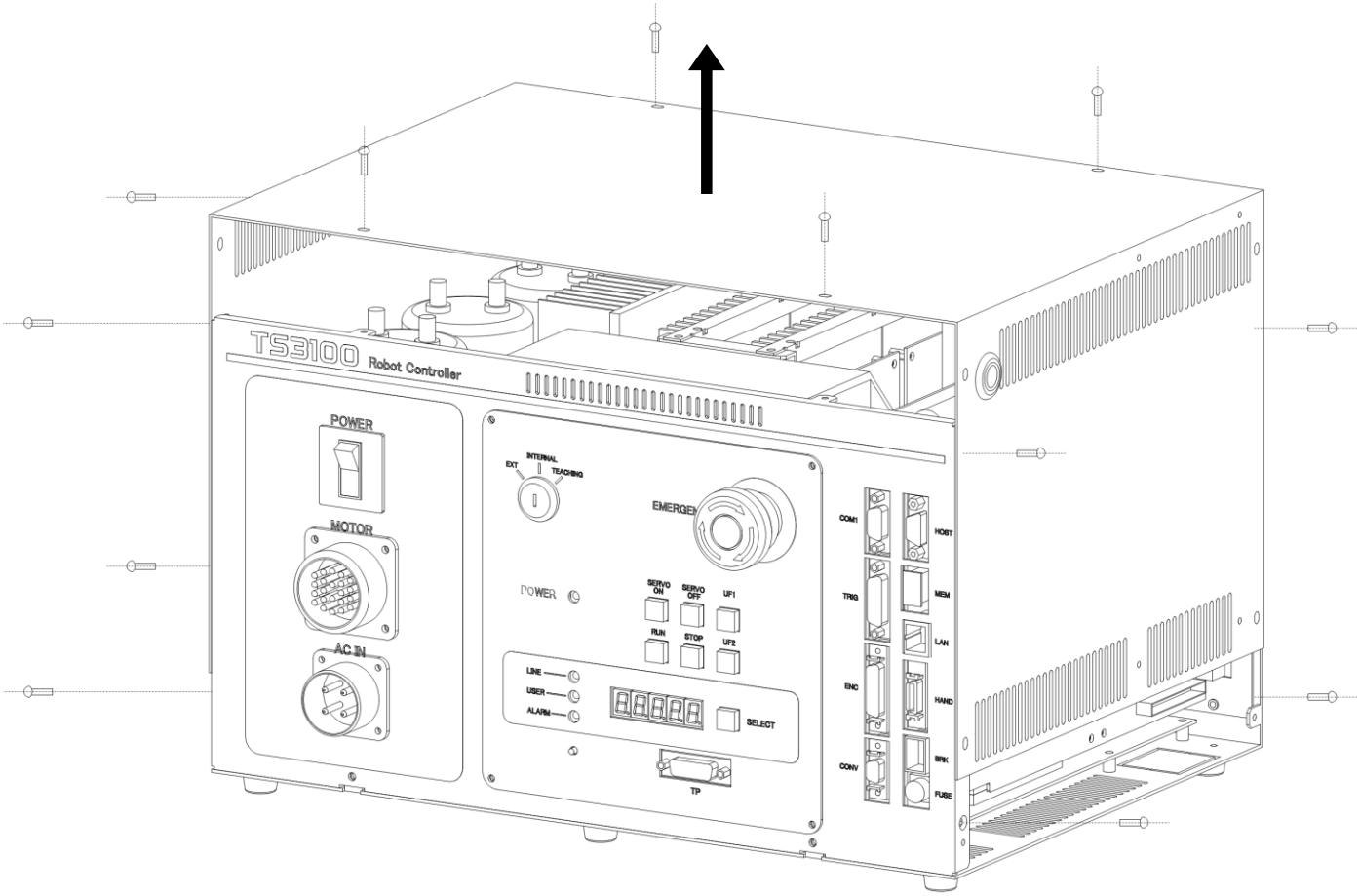


Fig. 3.9 Removal of upper cover (TS3100, TS3100E)

4) Remove the screws (two places) securing the switching power supply (PS1).

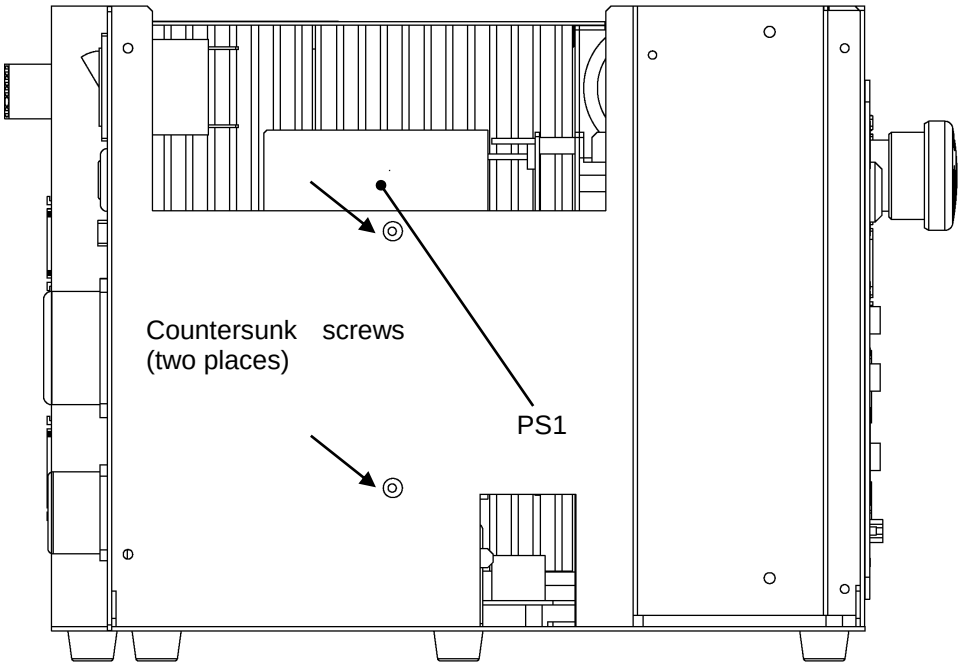


Fig. 3.10 Lower part of the left side (TS3000,TS3000E)

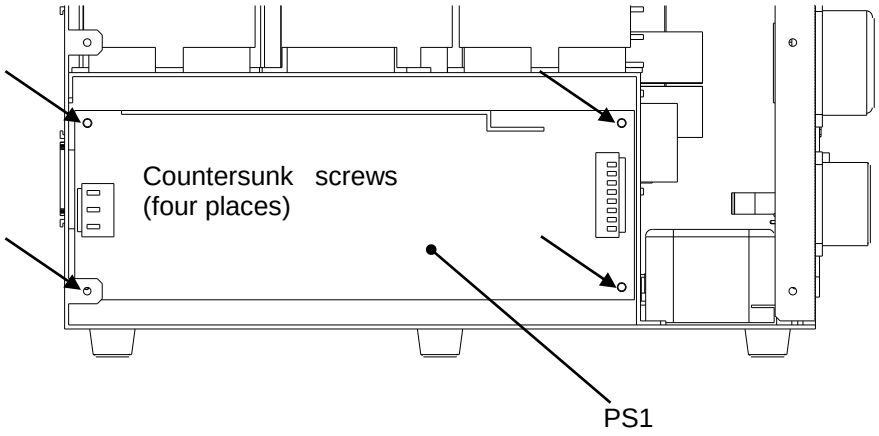


Fig. 3.11 Lower part of the left side (TS3100,TS3100E)

5) Secure the new power supply to the controller by tightening the screws.

6) Attach the cover to the controller.

Table 3.4 Type of PS1 switching power supply unit

Controller	Type	Maker
TS3000, TS3000E	PBA75F-24	COSEL
TS3100, TS3100E	LEB100F-0524	COSEL

Note: Only the battery shown above should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

[PS2 Replacement procedures]

1) Turn off the main power switch.
2) Turn off the power breaker.
3) Remove the cover from the controller.

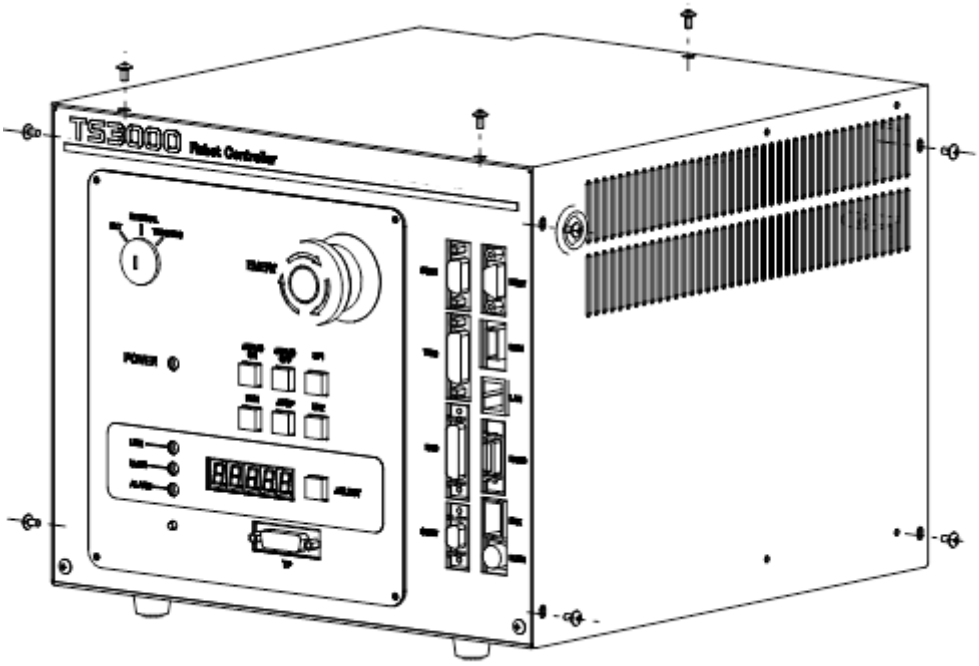


Fig. 3.12 Removal of upper cover (TS3000, TS3000E)

4) Remove the screws (two or three places) securing the servo driver.

Note: Mount servo driver in same position as before. When mounting mistake, the robot don't move.

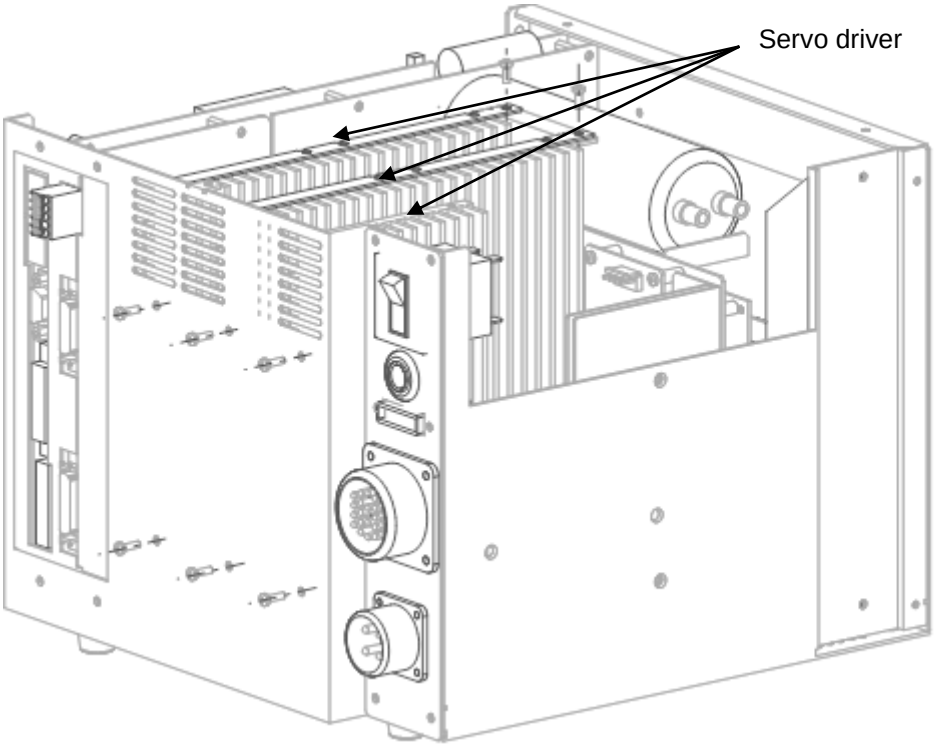


Fig. 3.13 Removal of servo driver (TS3000, TS3000E)

5) Remove the screws (four places) securing the switching power supply (PS2).

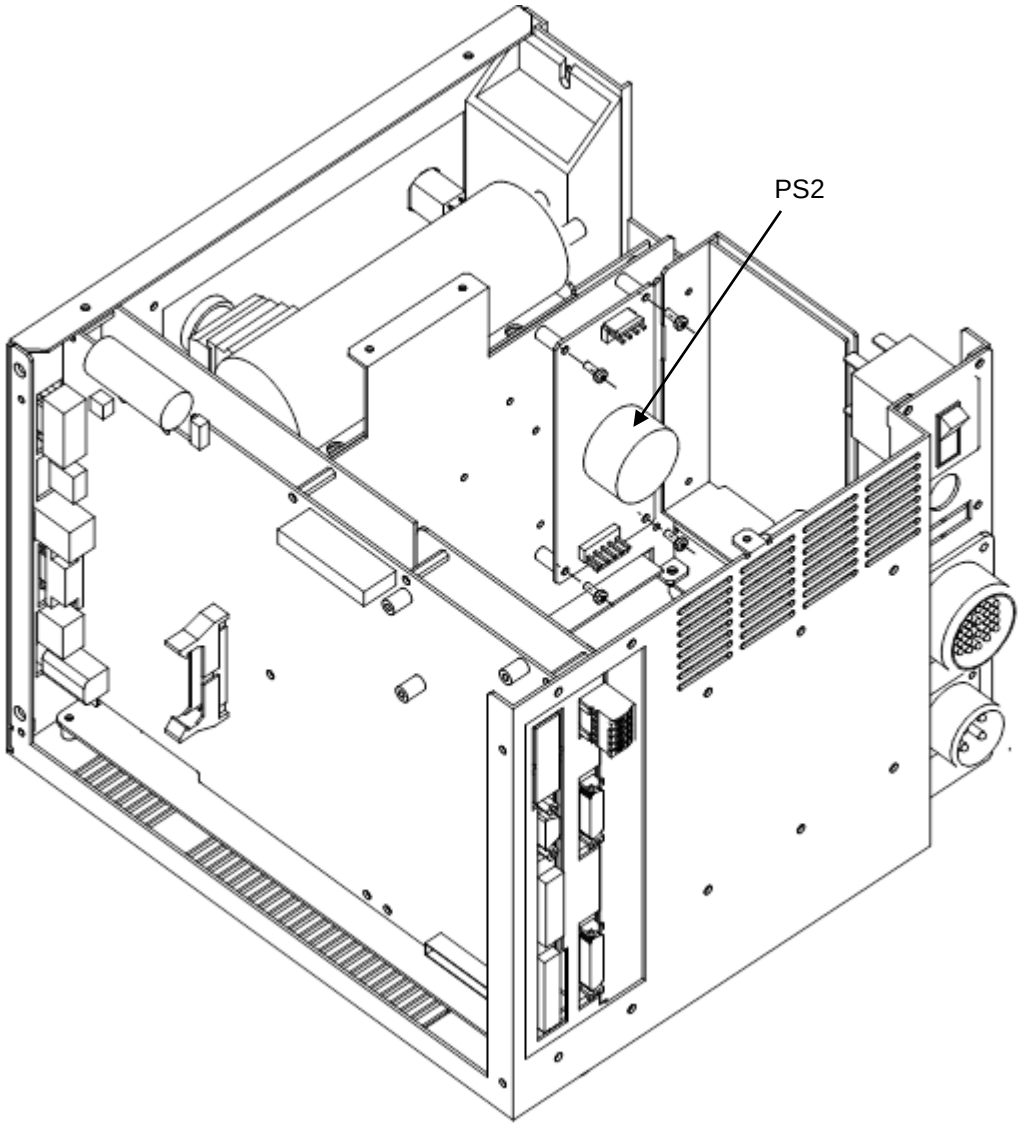


Fig. 3.14 Removal of PS2 switching power supply unit (TS3000, TS3000E)

6) Mount the new power supply unit.

7) Mount the cover to the main body.

Table 3.5 Type of PS2 switching power supply unit

Controller	Type	Maker
TS3000, TS3000E	LDA30F-5-Y	COSEL

Note: Only the battery shown above should be used. As this is an exclusive battery, contact SHIBAURA MACHINE at order entry.

3.4.5 Replacement of Fuse (X8GN, X8GI Printed Board)

If current exceeding the specified value flows into the I/O unit, the fuse on the front panel of the controller is blown. If the alarm saying "I/O FUSE Broken" (8–273) is displayed, replace the fuse with a new one.

If the fuse of the output module has been blown out, examine and identify an fault circuit, remedy the cause, then replace the fuse.

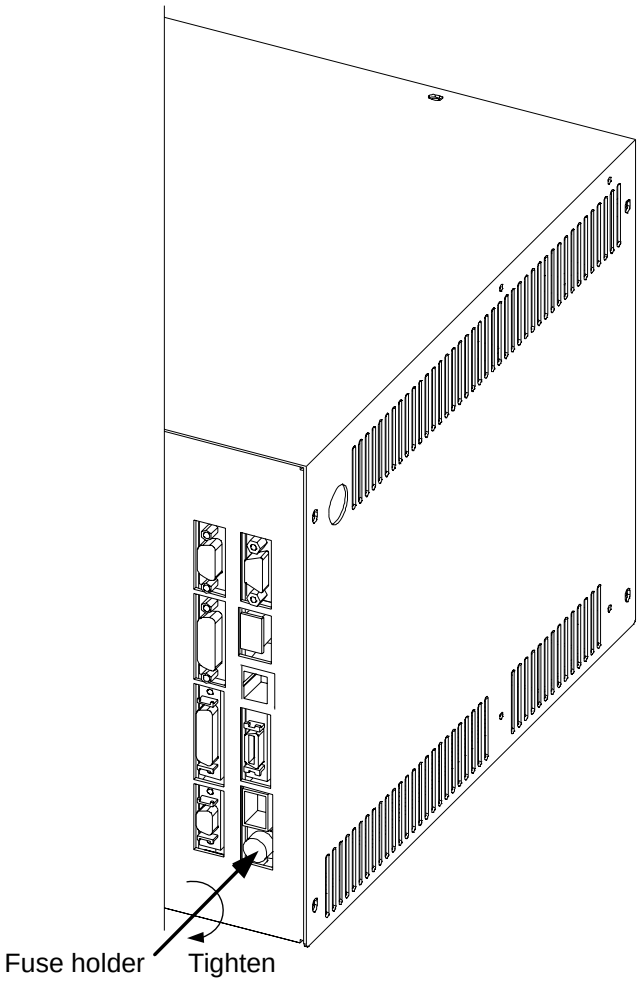


Fig. 3.15 Replacement of fuse

Table 3.6 Type of fuse

Type of fuse	Maker
51NM030H	PICO

[Replacement procedures]

1) Turn off the controller main power.
2) Remove the fuse holder in the above figure. (The holder lock is released by pushing and turning the holder 90° counterclockwise.)
3) Remove the fuse and set a new fuse.
4) Mount the fuse holder. (The holder is locked by pushing and turning the holder 90° counterclockwise.)
5) Turn on the controller main power and make sure that no error will occur.

3.4.6 Replacement of Output ICs (X8GC and X8GN (X8GI) Printed Boards)

If the current exceeding the specified current has run through the output unit, the outputs may not turn on normally.

If this happens, the ICs of the X8GC and X8GN (X8GI) printed boards will be damaged. Replace the ICs in that case.

Before the replacement, examine and identify an fault circuit, remedy the cause, then replace the ICs.

The following figure shows the relationships between the output ICs and signals:

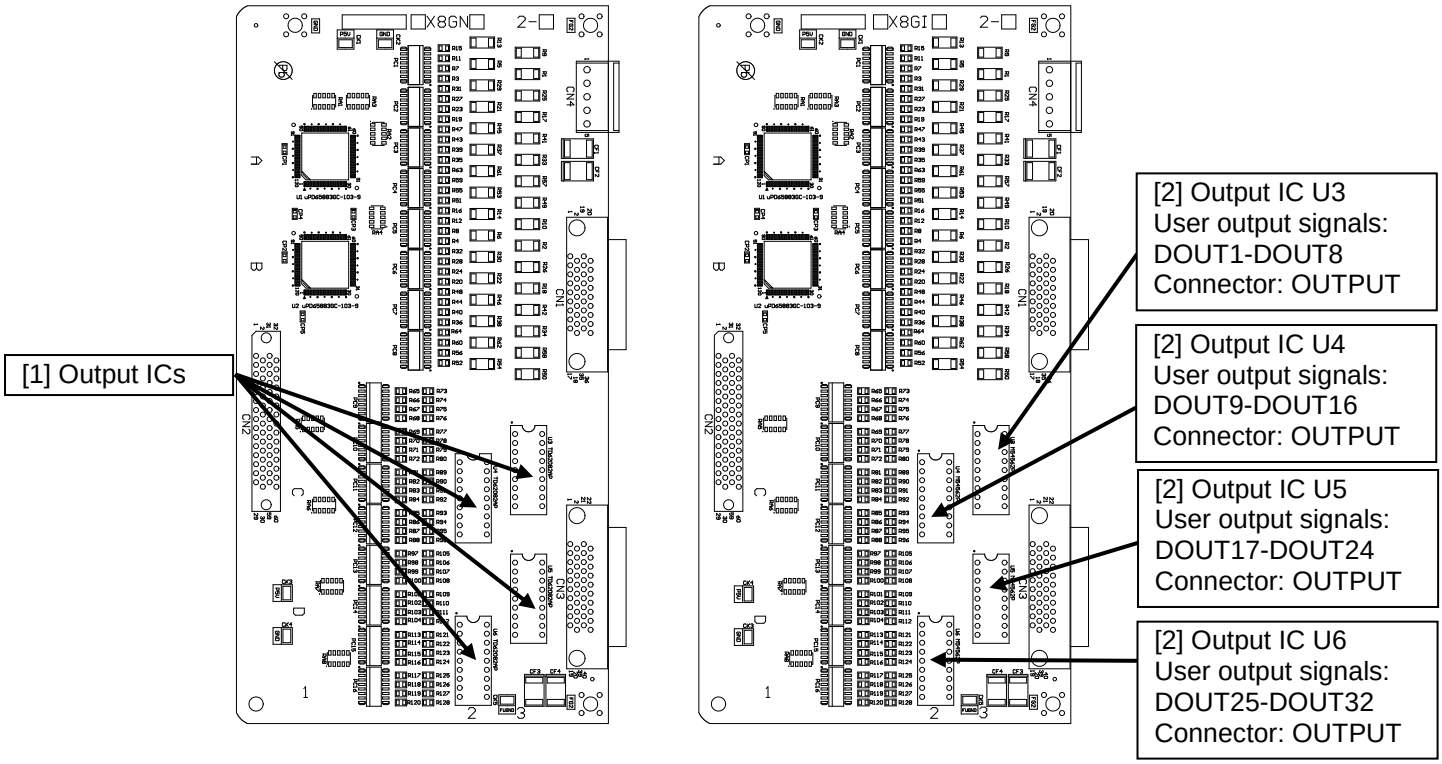


Fig. 3.16 I/O print circuit board (X8GN/X8GI)

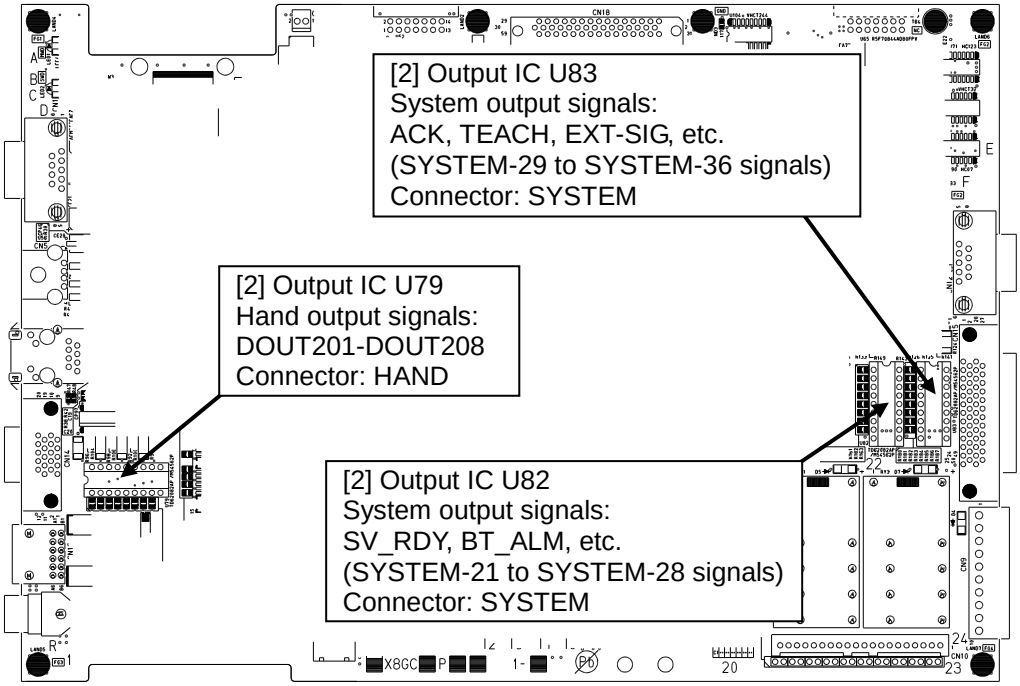


Fig. 3.17 Main printed circuit board (X8GC)

Table 3.7 Type of IC

	Name of board	Type of IC	Maker
[1]	X8GN	TD62082AP	Toshiba Corp.
[2]	X8GI	M54562P	Mitsubishi Electric

Replacement procedures

1) Turn off the controller main power.
2) Remove the cover from the body.
3) Remove all connectors in the control panel, which are linked to the X8GC and X8GN (X8GI) printed boards.
4) Remove the X8GC and X8GN (X8GI) printed boards from the body.
5) Remove the ICs shown in Fig 3.16 and fig.3.17 from respective sockets.
6) Attach new ICs to the sockets.
7) Mount the X8GC and X8GN (X8GI) printed boards on the body.
8) Link the removed connectors to the X8GC printed board, and set the main cover.
9) Turn on the controller main power.

4. Robot Home Point and Position Detector Error

4.1 Setting Robot Home Point

Before delivery from our plant, home point setting is performed for the robot after its arm is secured with clamp for home point setting.
At the time of home point setting, position data of the motor position detector (i.e., encoder) is backed up by batteries, and coordinates of the robot need not be set each time the power is connected.
The position data of the motor position detector comes in the two (2) types; Servo offset and multi-turn data.

Servo offset
At home point setting operation of the robot (i.e., ZEROP operation and REORG operation), the data are written into the parameter file. As the data is backed up by batteries, the parameters should be loaded again from the attached system disk at replacement of the main control printed board.

Multi-turn data:
This data is kept in the encoder by battery backup, which differs from the backup battery of the main control printed board. When the battery voltage drops, correct multi-turn data cannot be guaranteed and the encoder itself outputs an alarm. This data is set to zero (0) at home point setting operation of the robot (i.e., ZEROP operation and REORG operation). Likewise, it is set to zero (0) also by multi-turn data reset operation.

4.2 Position Detector Error

If a position detector error (encoder error) occurs, either of the following errors is shown on the error display of the robot controller.

Error code	Error content	Remarks
8–601	Axis1 Enc multi count err	
8–602	Axis2 Enc multi count err	
8–603	Axis3 Enc multi count err	
8–604	Axis4 Enc multi count err	
8–605	Axis5 Enc multi count err	Case of TS3100 or TS3100E (for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

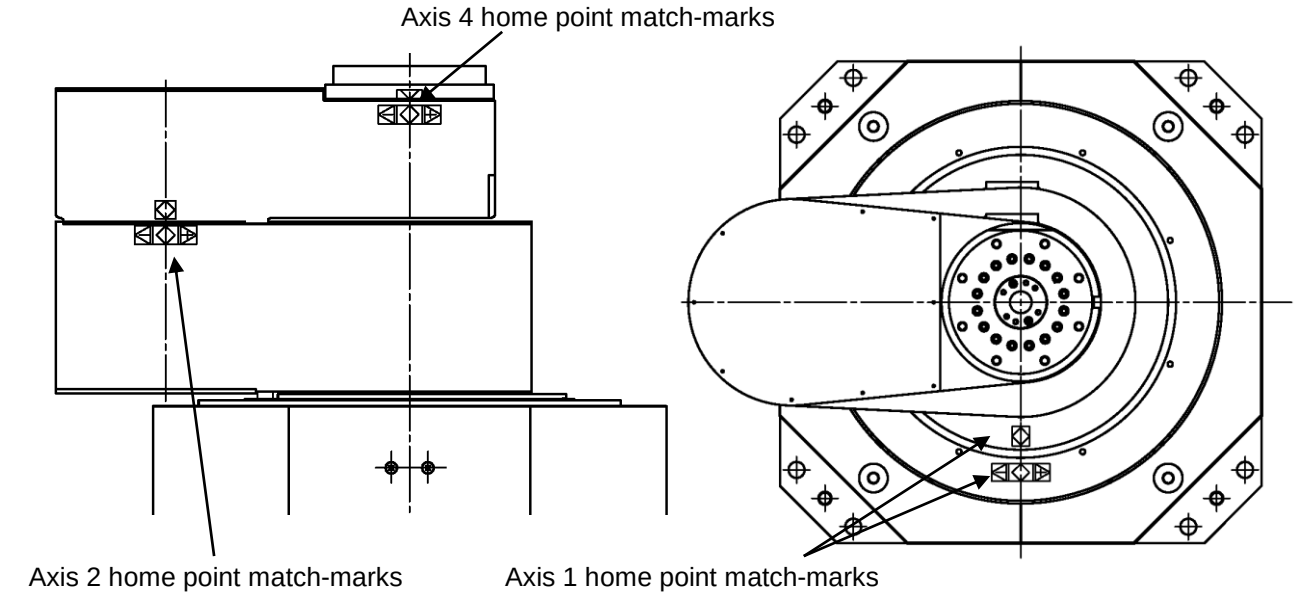
The position detector error is an error detected by the encoder itself. Even if the data in the parameter file is lost (or changed) to change the home point coordinates of the robot, for instance, a position detector error will not occur. If the machine coordinates have changed without causing a position detector error, the mechanical connecting position of the servo motor and mechanical unit shifted or the data in the parameter file described above was changed. (Ex.: Tooth skip of timing belt, etc.)

Errors detected by the encoder itself include the battery voltage drop, error caused by temperature rise in the encoder, counter overflow, internal counter data inconsistency, etc. Among these errors, the error that occurs most frequently is the battery voltage drop that is caused by the absence of maintenance at specified change intervals due to a long-term shutdown or cable breakage.
To prevent the position detector error, therefore, replace the batteries on a regular basis. For the battery replacement, see “Para. 2.3, Replacing Position Detector (or Encoder) Batteries”.
Also, a heavy vibration should not be exerted on the robot or the robot should not be moved suddenly by hand when the power is disconnected. Especially, when transporting and storing the robot, secure the robot in a posture for shipment, using the attached clamps.

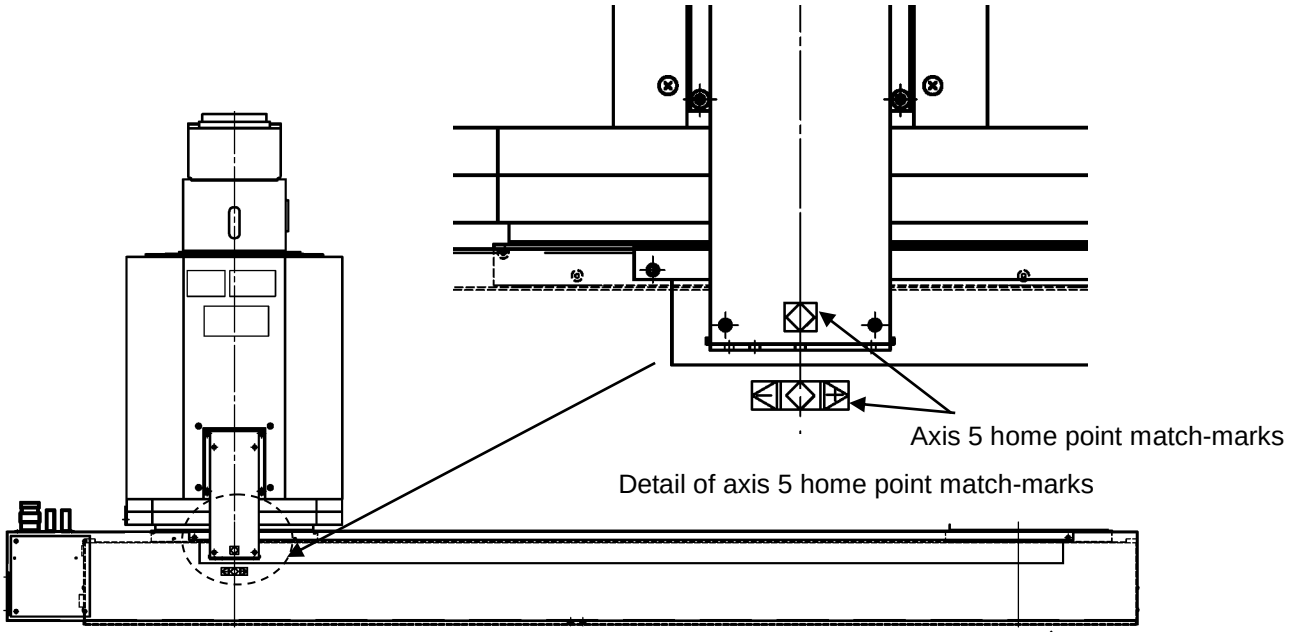
4.3 Procedure for Reproducing Home point Data at Battery Restoration

If the position detector batteries are left while the voltage has dropped, the position data of home point is lost. The following explains how to set home point if the home point data has been lost. The position detector batteries are used to save the multi-rotation data of the encoder. Thus, the position data of home point can be reproduced by visually aligning the target axis with the home point marks and resetting the multi-rotation data.

4.3.1 Reproducing the home point data of axes 1, 2, 4 (rotation) and 5 (traverse)



Locations of home point match-marks of axes 1, 2 and 4



Location of home point match-marks of axis 5 (only for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

Fig. 4.1 Home positions of axes 1, 2, 4 and 5

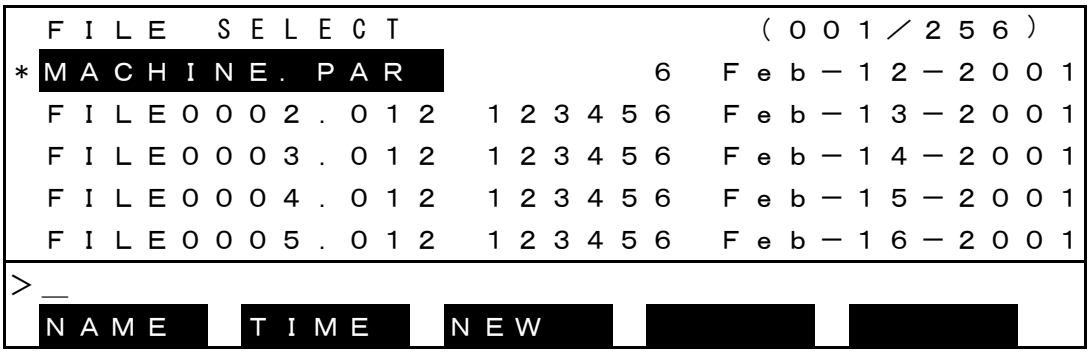
(1) Align the axis 1 with the home point match-marks. (See Fig. 4.1.)
(2) Turn off the servo, activate the operation mode [TEACH], and press [UTILITY] on the teach pendant.
(3) When the UTILITY screen appears, press the [NEXT] key once. [ENC] will be displayed. Press the function key [F5] corresponding to the displayed [ENC].
(4) When the encoder information screen appears, press the [ALT] and [0] keys together to release the edit lock. ([0]: Numeric key)
(5) Adjust the cursor to the axis whose multi-rotation data is to be reset (cleared to 0), and press the function key [F3] corresponding to [RESET].
(6) Move the robot to a teaching point, and check that the position is not shifted.
(7) To reproduce the home point data of the axes 2, 4 and 5, perform the same procedure as for the axis 1.

4.3.2 Reproducing the zero-point data of axis 3 (Z)

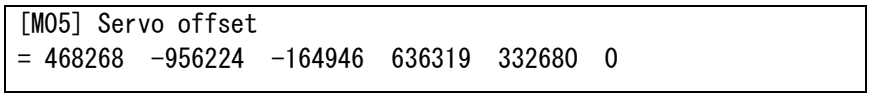
The motor brake is applied to the axis 3, so it is impossible for operators to align the axis with the home point match-marks. However, the position data can be reproduced by the following method: setting the current position as a temporary home point, adjusting the axis to the specified height in JOG mode, and specifying the pre-recorded encoder offset default value in the system file (MACHINE.PAR).

a) Record the encoder offset default value onto clean paper or the like.

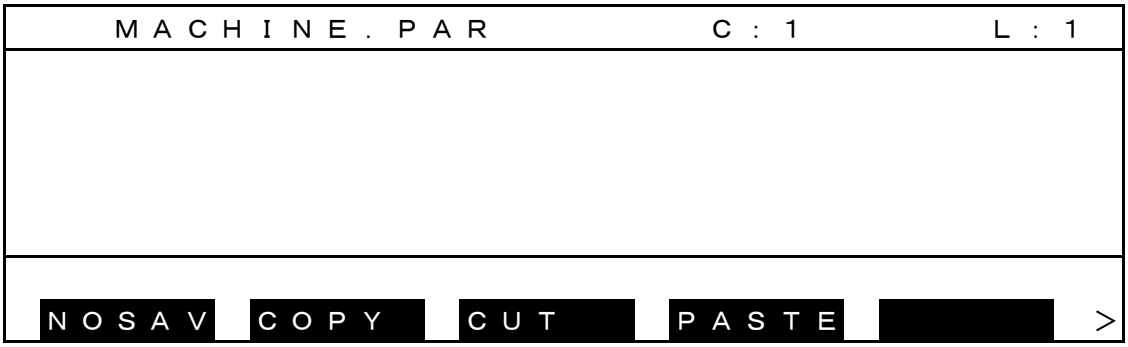
1) Activate the operation mode [TEACH] when the servo is off, open the machine parameter file "**MACHINE.PAR**" through [EDIT] on the teach pendant. The "**MACHINE.PAR**" file contains the settings of machine-specific data such as home points and options.



2) View [M05] Servo offset shown below. The third value from left ("-164946" in the following example) is the encoder offset value of the axis 3. Record the value onto clean paper or the like. Be sure to record the value because it is overwritten with a different value when the temporary zero point is set.

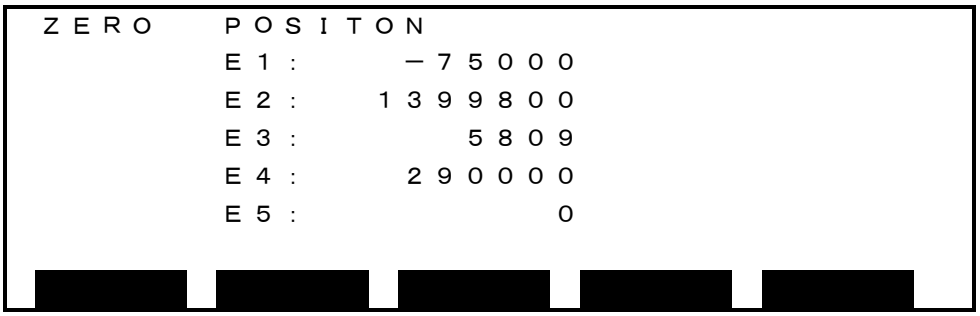


3) Close the machine parameter file "**MACHINE.PAR**" by executing [NOSAV], but not executing SAVE.

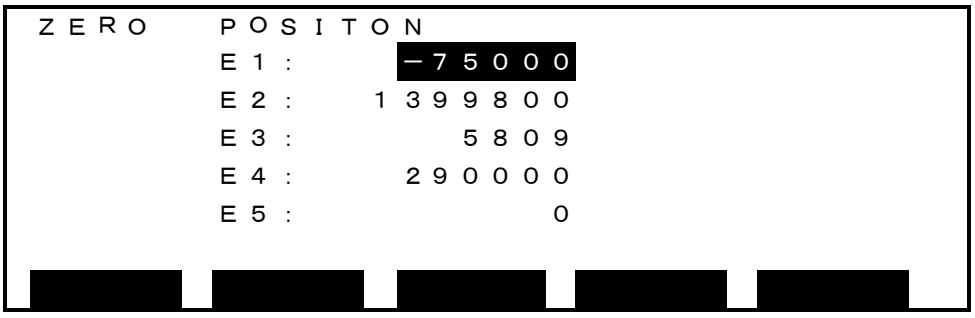


b) Set the current position as a temporary home point (using the teach pendant).

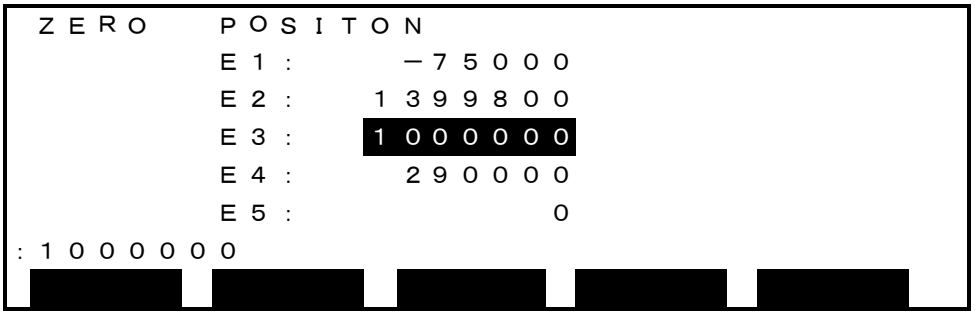
1) Select the TEACHING mode as the master mode.
2) Press the [UTILITY] key on the teach pendant.
3) Press the [NEXT] key (F6) on the teach pendant twice.
4) Turn off the servo power.
5) Press the [F3] key on the teach pendant to select the ZEROP mode.
6) The following home point setting screen is called.



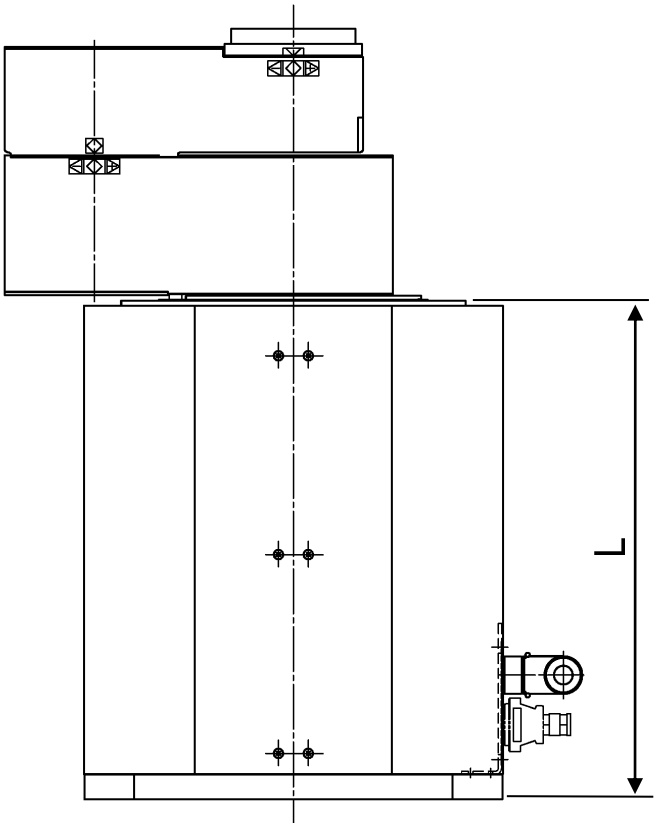
7) Press the [ALT] and [0] keys together on the teach pendant.
8) The following home point editing screen appears.
9) The value of E1 (axis 1) is highlighted.



10) Press the [↓] key on the teach pendant twice. The value of E3 (axis 3) will be highlighted.
11) Press the [EXE] key on the teach pendant, and enter "***0000" as the value of E3. *** is the axis-3 (Z) stroke. For example, if the stroke is 100 mm, enter "1000000."



c) Slowly move the axis 3 down to near the lower limit, and adjust the dimension "L" in Fig. 4.2 to the value shown in the table. (Dimension L: Distance from the robot shoulder to the bottom of the base)



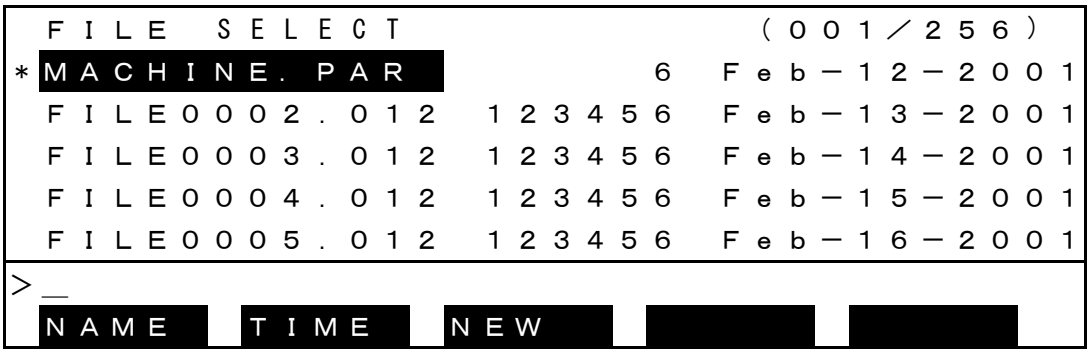
Axis 3 (Z) stroke (mm)	100	150
L (mm) *1	502	562

*1 The dimension of the robot may vary depending on the robot specifications. Be sure to see the external view of the robot attached to the specifications.

Fig. 4.2 Dimension L of the axis 3 (Z)

d) Enter the encoder offset default value (using the teach pendant).

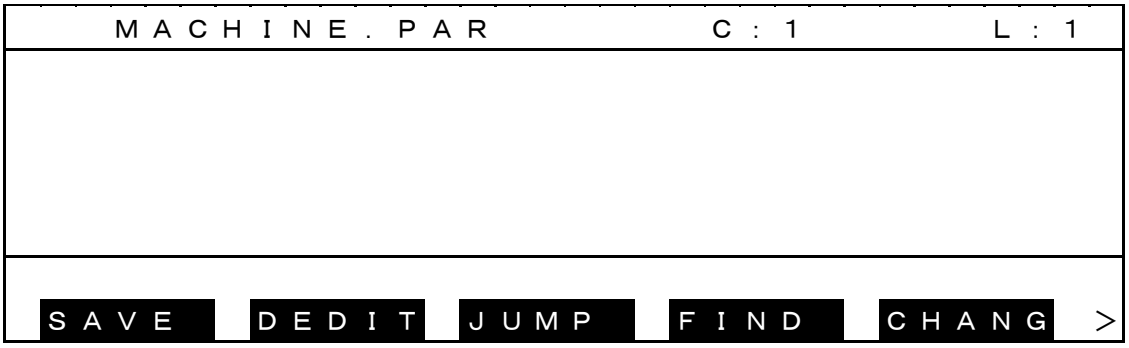
1) Activate the operation mode [TEACH] when the servo is off, open the machine parameter file "**MACHINE.PAR**" again through [EDIT] on the teach pendant.



2) Change the setting of [M05] Servo offset shown below. The third value from left ("-164946" in the following example) is the encoder offset value of the axis 3. Change the value to the one you recorded in Item a).

```
[M05] Servo offset
= 468268  -956224  -164946  636319  332680  0
```

3) Execute [SAVE] to save the machine parameter file "MACHINE.PAR" and close it.



e) Turn off the controller, and then turn it on. Finally, move the robot to a teaching point, and check that the position is not shifted.

This completes the setting.

5. Replacement Parts for Maintenance

5.1 Cautions for maintenance replacement parts



CAUTION

- With the exception of the greases, all parts listed in List of Replacement Parts-Main Robot are custom-made to SHIBAURA MACHINE specifications. Contact SHIBAURA MACHINE at order entry.
- Parts replacement work must be done by SHIBAURA MACHINE service engineer.
- SHIBAURA MACHINE's warranty does not cover failures, accidents or any damages.

5.2 List of Replacement Parts – Main Robot (SR-404HC)

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor	VLBST-Z06030S-OU001		Y610A1D90	SHIBAURA MACHINE	1	Axis 1
2		VLBST-Y04030S-OU001		Y610A1D50		2	Axes 2,4
3		VLBST-Y04030S-BOU004		Y610A5630		1	Axis 3
4	Reduction gear		S694718		SHIBAURA MACHINE	1	Axis 1
5			368M7202			1	Axis 2
6			368M7102			1	Axis 4
7	Timing pulley		S727312 (On the drive side)		SHIBAURA MACHINE	1	Axis 3
8			S727313 (On the driven side)			1	Axis 3
9			S727314 (On the drive side in the base)			2	Axes 2,4
10			S727315 (On the driven side in the base)			1	Axis 4
11			S727316 (On the driven side in the base)			1	Axis 2
12			S727303 (On the drive side in the arm 1)			1	Axis 2
13			S727384 (On the driven side in the arm1)			1	Axis 2
14			S727385 (On the drive side in the arm 2)			1	Axis 4
15			S727386 (On the driven side in the arm2)			1	Axis 4
16			S700122 (On the drive side in the arm 1)			1	Axis 4
17			S720382 (On the driven side in the arm1)			1	Axis 4
18	Timing belt	513-3GT-12			SHIBAURA MACHINE	1	Axis 3
19		624-3GT-12 (In the arm)				1	Axis 2
20		408-3GT-12 (In the base)				2	Axes 2,4
21		624-3GT-12 (In the arm)				2	Axis 4
22	Ball screw		M060979		SHIBAURA MACHINE	1	Axis 3
23	Linear guide		M067824		SHIBAURA MACHINE	2	Axis 3
24	Support unit	FK20			SHIBAURA MACHINE	1	Axis 3
25	Encoder backup battery		KK1613G01 (Lithium battery)		SHIBAURA MACHINE	2	All axes
26	Grease	Moly-White grease RE No.00			Nabtesco		Axis 1
27		AFE grease			THK		Axis 3

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
28			KK7628G01 (Power cable)			1	CN2
29			M170225 (Encoder cable)			1	CN3
30			M170226 (Axis-3 brake cable)			1	CN4
31	Harness		KL3219G01 (Hand I/O)		SHIBAURA	1	
32	*1		KL1665G01 (Hand I/O)		MACHINE	1	In the arm
33			339N2701 (Robot-controller cable: 5 m)			1	CN2
34			339N2801 (Robot-controller cable: 5 m)			1	CN3
35			339N2901 (Robot-controller cable: 5 m)			1	CN4

*1 For details, see Fig 5.1 Component Connection Diagram.

5.3 List of Replacement Parts – Main Robot (SR-404HC-Z150)

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor	VLBST-Z06030S-OU001		Y610A1D90	SHIBAURA MACHINE	1	Axis 1
2		VLBST-Y04030S-OU001		Y610A1D50		2	Axes 2,4
3		VLBST-Y04030S-BOU004		Y610A5630		1	Axis 3
4	Reduction gear		S694718		SHIBAURA MACHINE	1	Axis 1
5			368M7202			1	Axis 2
6			368M7102			1	Axis 4
7	Timing pulley		S727312 (On the drive side)		SHIBAURA MACHINE	1	Axis 3
8			S727313 (On the driven side)			1	Axis 3
9			S727314 (On the drive side in the base)			2	Axes 2,4
10			S727315 (On the driven side in the base)			1	Axis 4
11			S727316 (On the driven side in the base)			1	Axis 2
12			S727303 (On the drive side in the arm 1)			1	Axis 2
13			S727384 (On the driven side in the arm1)			1	Axis 2
14			S727385 (On the drive side in the arm 2)			1	Axis 4
15			S727386 (On the driven side in the arm2)			1	Axis 4
16			S700122 (On the drive side in the arm 1)			1	Axis 4
17			S720382 (On the driven side in the arm1)			1	Axis 4
18	Timing belt	513-3GT-12			SHIBAURA MACHINE	1	Axis 3
19		624-3GT-12 (In the arm)				1	Axis 2
20		408-3GT-12 (In the base)				2	Axes 2,4
21		624-3GT-12 (In the arm)				2	Axis 4
22	Ball screw		M243197 (For Z150)		SHIBAURA MACHINE	1	Axis 3
23	Linear guide		M243198 (For Z150)		SHIBAURA MACHINE	2	Axis 3
24	Support unit	FK20			SHIBAURA MACHINE	1	Axis 3
25	Encoder backup battery		KK1613G01 (Lithium battery)		SHIBAURA MACHINE	2	All axes
26	Grease	Moly-White grease RE No.00			Nabtesco		Axis 1
27		AFE grease			THK		Axis 3

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
28			M243233 (Power cable for Z150)			1	CN2
29			M243234 (Encoder cable for Z150)			1	CN3
30			M170226 (Axis-3 brake cable)			1	CN4
31	Harness		KL3219G01 (Hand I/O)		SHIBAURA MACHINE	1	
32	*1		KL1665G01 (Hand I/O)			1	In the arm
33			339N2701 (Robot-controller cable: 5 m)			1	CN2
34			339N2801 (Robot-controller cable: 5 m)			1	CN3
35			339N2901 (Robot-controller cable: 5 m)			1	CN4

*1 For details, see Fig 5.1 Component Connection Diagram.

5.4 List of Replacement Parts – Main Robot (SR-624HC)

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor	VLBST-Z06030S-OU001		Y610A1D90	SHIBAURA MACHINE	1	Axis 1
2		VLBST-Y04030S-OU001		Y610A1D50		2	Axes 2,4
3		VLBST-Y04030S-BOU004		Y610A5630		1	Axis 3
4	Reduction gear		S694718		SHIBAURA MACHINE	1	Axis 1
5			368M7302			1	Axis 2
6			368M7202			1	Axis 4
7	Timing pulley		S727312 (On the drive side)		SHIBAURA MACHINE	1	Axis 3
8			S727313 (On the driven side)			1	Axis 3
9			S727314 (On the drive side in the base)			2	Axes 2,4
10			S727315 (On the driven side in the base)			1	Axis 4
11			S727316 (On the driven side in the base)			1	Axis 2
12			S727303 (On the drive side in the arm 1)			1	Axis 2
13			S727306 (On the driven side in the arm1)			1	Axis 2
14			S727307 (On the drive side in the arm 2)			1	Axis 4
15			S727308 (On the driven side in the arm2)			1	Axis 4
16			S727304 (On the drive side in the arm 1)			1	Axis 4
17			S727305 (On the driven side in the arm1)			1	Axis 4
18	Timing belt	513-3GT-12			SHIBAURA MACHINE	1	Axis 3
19		840-3GT-12 (In the arm)				1	Axis 2
20		408-3GT-12 (In the base)				2	Axes 2,4
21		840-3GT-12 (In the arm)				2	Axis 4
22	Ball screw		M060979		SHIBAURA MACHINE	1	Axis 3
23	Linear guide		M067824		SHIBAURA MACHINE	2	Axis 3
24	Support unit	FK20			SHIBAURA MACHINE	1	Axis 3
25	Encoder backup battery		KK1613G01 (Lithium battery)		SHIBAURA MACHINE	2	All axes
26	Grease	Moly-White grease RE No.00			Nabtesco		Axis 1
27		AFE grease			THK		Axis 3

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
28	Harness *1		KK7628G01 (Power cable)		SHIBAURA MACHINE	1	CN2
29			M170225 (Encoder cable)			1	CN3
30			M170226 (Axis-3 brake cable)			1	CN4
31			339N2701 (Robot-controller cable: 5 m)			1	CN2
32			339N2801 (Robot-controller cable: 5 m)			1	CN3
33			339N2901 (Robot-controller cable: 5 m)			1	CN4

*1 For details, see Fig 5.2 Component Connection Diagram.

5.5 List of Replacement Parts – Main Robot (SR-405HC-L1550-R/L)

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor	VLBST-Z06030S-OU001		Y610A1D90	SHIBAURA MACHINE	1	Axis 1
2		VLBST-Y04030S-OU001		Y610A1D50		2	Axes 2,4
3		VLBST-Y04030S-BOU004		Y610A5630		1	Axis 3
4		VLBST-Z08030S-OU001		Y610A1D20		1	Axis 5
5	Reduction gear		S694718		SHIBAURA MACHINE	1	Axis 1
6			368M7202			1	Axis 2
7			368M7102			1	Axis 4
8	Timing pulley		S727312 (On the drive side)		SHIBAURA MACHINE	1	Axis 3
9			S727313 (On the driven side)			1	Axis 3
10			S727314 (On the drive side in the base)			2	Axes 2,4
11			S727315 (On the driven side in the base)			1	Axis 4
12			S727316 (On the driven side in the base)			1	Axis 2
13			S727303 (On the drive side in the arm 1)			1	Axis 2
14			S727384 (On the driven side in the arm1)			1	Axis 2
15			S727385 (On the drive side in the arm 2)			1	Axis 4
16			S727386 (On the driven side in the arm2)			1	Axis 4
17			S700122 (On the drive side in the arm 1)			1	Axis 4
18			S720382 (On the driven side in the arm1)			1	Axis 4
19	Timing belt	513-3GT-12			SHIBAURA MACHINE	1	Axis 3
20		624-3GT-12 (In the arm)				1	Axis 2
21		408-3GT-12 (In the base)				2	Axes 2,4
22		624-3GT-12 (In the arm)				2	Axis 4
23	Ball screw		M060979		SHIBAURA MACHINE	1	Axis 3
24			T-L2549503			1	Axis 5
25	Linear guide		M067824		SHIBAURA MACHINE	2	Axis 3
26			3BV0297001			2	Axis 5
27	Support unit	FK20			SHIBAURA MACHINE	1	Axis 3
28	Encoder backup battery		KK1613G01 (Lithium battery)		SHIBAURA MACHINE	2	All axes
29	Grease	Moly-White grease RE No.00			Nabtesco		Axis 1
30		AFE grease			THK		Axis 3
31		LG2			NSK		Axis 5

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
32			KK7628G01 (Power cable)			1	CN2
33			M170225 (Encoder cable)			1	CN3
34			M170226 (Axis-3 brake cable)			1	CN4
35			KL3219G01 (Hand I/O)			1	
36			KL1665G01 (Hand I/O)			1	Inthearm
37	Harness		300N4401 (In the traverse axis)		SHIBAURA MACHINE	1	CN2
38	*1		300N4501 (In the traverse axis)			1	CN3
39			300N4601 (In the traverse axis)			1	CN4
40			300N4701 (In the traverse axis)			1	CN11
41			396N3501 (Robot-controller cable: 5 m)			1	CN2
42			396N3601 (Robot-controller cable: 5 m)			1	CN3
43			396N3701 (Robot-controller cable: 5 m)			1	CN4

*1 For details, see Fig 5.3 Component Connection Diagram.

5.6 List of Replacement Parts – Main Robot (SR-405HC-L1530S-R/L)

No	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
1	AC servo motor	VLBST-Z06030S-OU001		Y610A1D90	SHIBAURA MACHINE	1	Axis 1
2		VLBST-Y04030S-OU001		Y610A1D50		2	Axes 2,4
3		VLBST-Y04030S-BOU004		Y610A5630		1	Axis 3
4		VLBST-Z08030S-OU001		Y610A1D20		1	Axis 5
5	Reduction gear		S694718		SHIBAURA MACHINE	1	Axis 1
6			368M7202			1	Axis 2
7			368M7102			1	Axis 4
8	Timing pulley		S727312 (On the drive side)		SHIBAURA MACHINE	1	Axis 3
9			S727313 (On the driven side)			1	Axis 3
10			S727314 (On the drive side in the base)			2	Axes 2,4
11			S727315 (On the driven side in the base)			1	Axis 4
12			S727316 (On the driven side in the base)			1	Axis 2
13			S727303 (On the drive side in the arm 1)			1	Axis 2
14			S727384 (On the driven side in the arm1)			1	Axis 2
15			S727385 (On the drive side in the arm 2)			1	Axis 4
16			S727386 (On the driven side in the arm2)			1	Axis 4
17			S700122 (On the drive side in the arm 1)			1	Axis 4
18			S720382 (On the driven side in the arm1)			1	Axis 4
19	Timing belt	513-3GT-12			SHIBAURA MACHINE	1	Axis 3
20		624-3GT-12 (In the arm)				1	Axis 2
21		408-3GT-12 (In the base)				2	Axes 2,4
22		624-3GT-12 (In the arm)				2	Axis 4
23	Ball screw		M060979		SHIBAURA MACHINE	1	Axis 3
24			M342196 (For L1530S)			1	Axis 5
25	Linear guide		M067824		SHIBAURA MACHINE	2	Axis 3
26			3BV0297001			2	Axis 5
27	Support unit	FK20			SHIBAURA MACHINE	1	Axis 3
28	Encoder backup battery		KK1613G01 (Lithium battery)		SHIBAURA MACHINE	2	All axes
29	Grease	Moly-White grease RE No.00			Nabtesco		Axis 1
30		AFE grease			THK		Axis 3
31		LG2			NSK		Axis 5

No.	Part name	Type	Dwg. No.	Unit code	Maker	Q'ty	Remarks
32			KK7628G01 (Power cable)			1	CN2
33			M170225 (Encoder cable)			1	CN3
34			M170226 (Axis-3 brake cable)			1	CN4
35			KL3219G01 (Hand I/O)			1	
36			KL1665G01 (Hand I/O)			1	Inthearm
37	Harness		300N4401 (In the traverse axis)		SHIBAURA MACHINE	1	CN2
38	*1		300N4501 (In the traverse axis)			1	CN3
39			300N4601 (In the traverse axis)			1	CN4
40			300N4701 (In the traverse axis)			1	CN11
41			396N3501 (Robot-controller cable: 5 m)			1	CN2
42			396N3601 (Robot-controller cable: 5 m)			1	CN3
43			396N3701 (Robot-controller cable: 5 m)			1	CN4

*1 For details, see Fig 5.3 Component Connection Diagram.

5.7 List of Replacement Parts – Controller (TS3000, TS3000E)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	PBA75F-24		COSEL	1	
2	PS2 (Switching power supply)	LDA30F-5-Y			1	
3	Main power switch	IR-11-A8-10-1BF		Nippon Thermo	1	
4	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	
5	Fuse	51NM030H		PICO	1	For X8GC printed boards
6	Printed board	X8GCA (Main control)	Y610A4000	SHIBAURA MACHINE	1	Type N
7		X8GCC (Main control)	Y610A4010		1	Type N, CE/Category3 (For TS3000E)
8		X8GCA (Main control)	Y610A4020		1	Type P
9		X8GCC (Main control)	Y610A4030		1	Type P, CE/Category3 (For TS3000E)
10		X8GLA (Servo logic)	Y610A4040		1	
11		X8GNA (I/O board)	Y610A40A0		1	Type N
12		X8GIA (I/O board)	Y610A40B0		1	Type P
13		X8G2A (20A) (Servo)	Y610A40F0		1	Axis 1
14		X8G2B (15A) (Servo)	Y610A40G0		3	Axes 2, 3, 4
15		X8GHA (Servo power supply)	Y610A40S0		1	
16		X8GBA (Mother board)	Y610A40R0		1	
17	Operation panel unit	X8GOA	Y610A4080	SHIBAURA MACHINE	1	Standard
18		X8GOD	Y610A40U0		1	CE/Category3 (For TS3000E)
19	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
20		TP1000 remodeling CE	Y610A2250		1	CE/Category3 (For TS3000E)
21	System disk	TS3000SYS	Y610D0740	SHIBAURA MACHINE	1	DVD-R
22	Replacement of Output ICs	TD62084APG		Toshiba	1	Type N
23		M54562WP		Mitsubishi Electric	1	Type P

5.8 List of Replacement Parts – Controller (TS3100, TS3100E)

No.	Part name	Type	Unit code	Maker	Q'ty	Remarks
1	PS1 (Switching power supply)	PBA75F-24		COSEL	1	
2	Main power switch	IR-11-A8-10-1BF		Nippon Thermo	1	
3	Lithium battery	ER6V C4 (ER6C WK27 was obsolete)		TOSHIBA	1	
4	Fuse	51NM030H		PICO	1	For X8GC printed boards
5	Printed board	X8GCA (Main control)	Y610A4000	SHIBAURA MACHINE	1	Type N
6		X8GCC (Main control)	Y610A4010		1	Type N, CE/Category3 (For TS3100E)
7		X8GCA (Main control)	Y610A4020		1	Type P
8		X8GCC (Main control)	Y610A4030		1	Type P, CE/Category3 (For TS3100E)
9		X8GLA (Servo logic)	Y610A4040		1	
10		X8GNA (I/O board)	Y610A40A0		1	Type N
11		X8GIA (I/O board)	Y610A40B0		1	Type P
12		X8G1C (25A) (Servo)	Y610A40E0		1	Axis 5
13		X8G2A (20A) (Servo)	Y610A40F0		1	Axis 1
14		X8G2B (15A) (Servo)	Y610A40G0		3	Axes 2, 3, 4
15		X8GHA (Servo power supply)	Y610A40S0		1	
16		X8GBA (Mother board)	Y610A40R0		1	
17	Operation panel unit	X8GOA	Y610A4080	SHIBAURA MACHINE	1	Standard
18		X8GOD	Y610A40U0		1	CE/Category3 (For TS3000E)
19	Teach pendant	TP1000	Y610A2600	SHIBAURA MACHINE	1	With 5 m-long cable
20		TP1000 remodeling CE	Y610A2250		1	CE/Category3 (For TS3000E)
21	System disk	TS3000SYS	Y610D0740	SHIBAURA MACHINE	1	DVD-R
22	Replacement of Output ICs	TD62084APG		Toshiba	1	Type N
23		M54562WP		Mitsubishi Electric	1	Type P

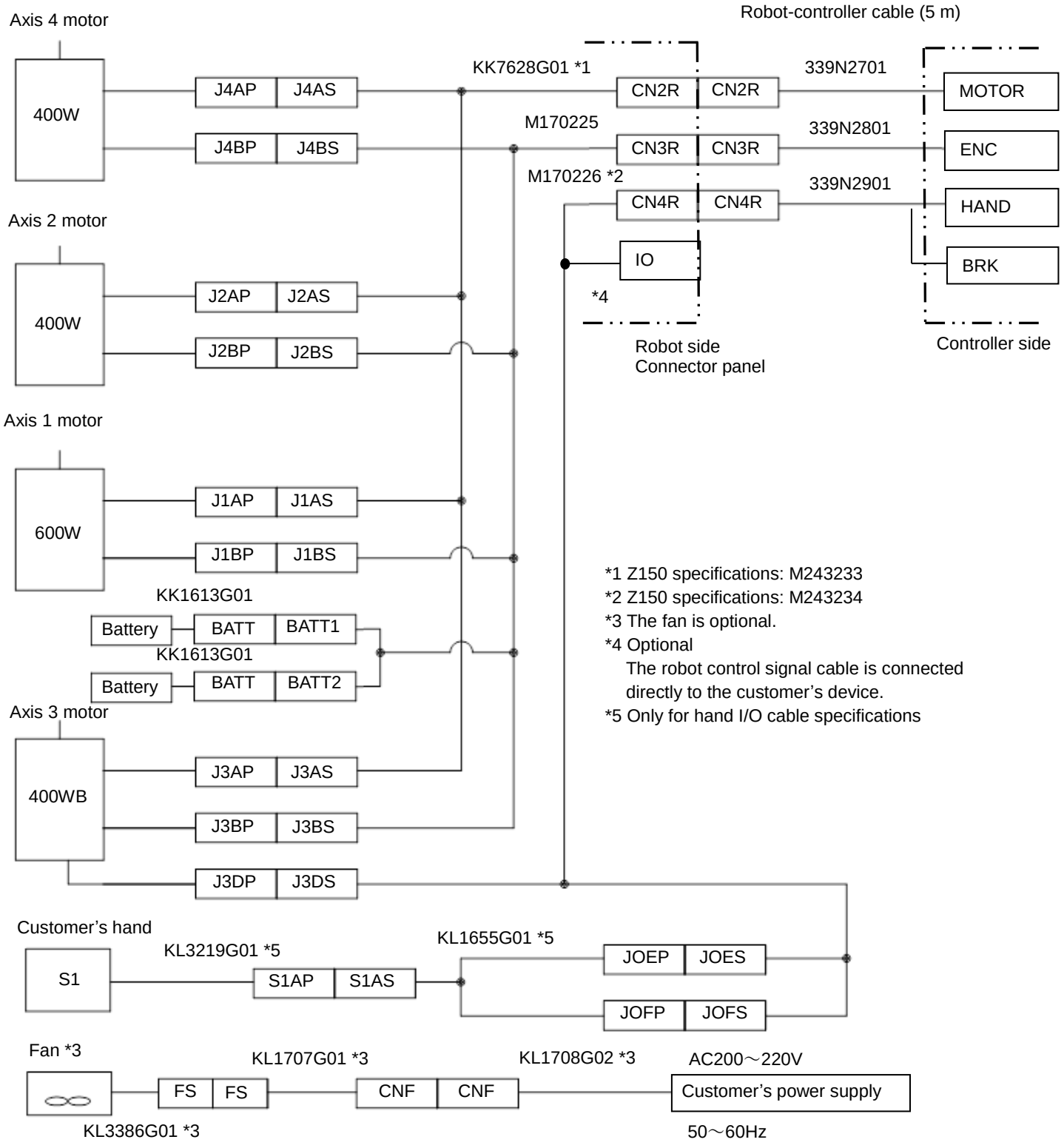


Fig. 5.1 Component Connection Diagram (SR-404HC and SR-404HC-Z150)

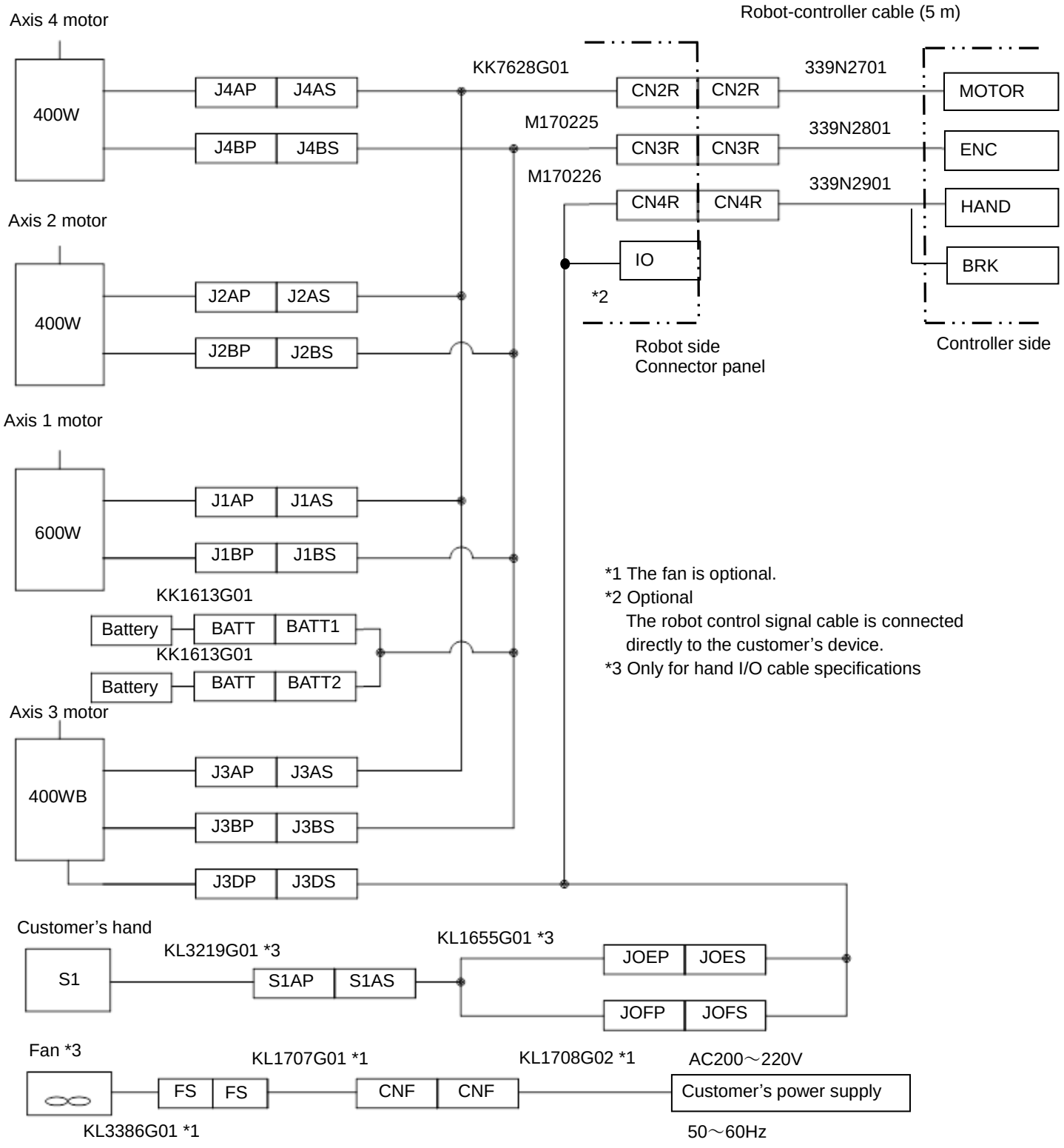


Fig. 5.2 Component Connection Diagram (SR-624HC)

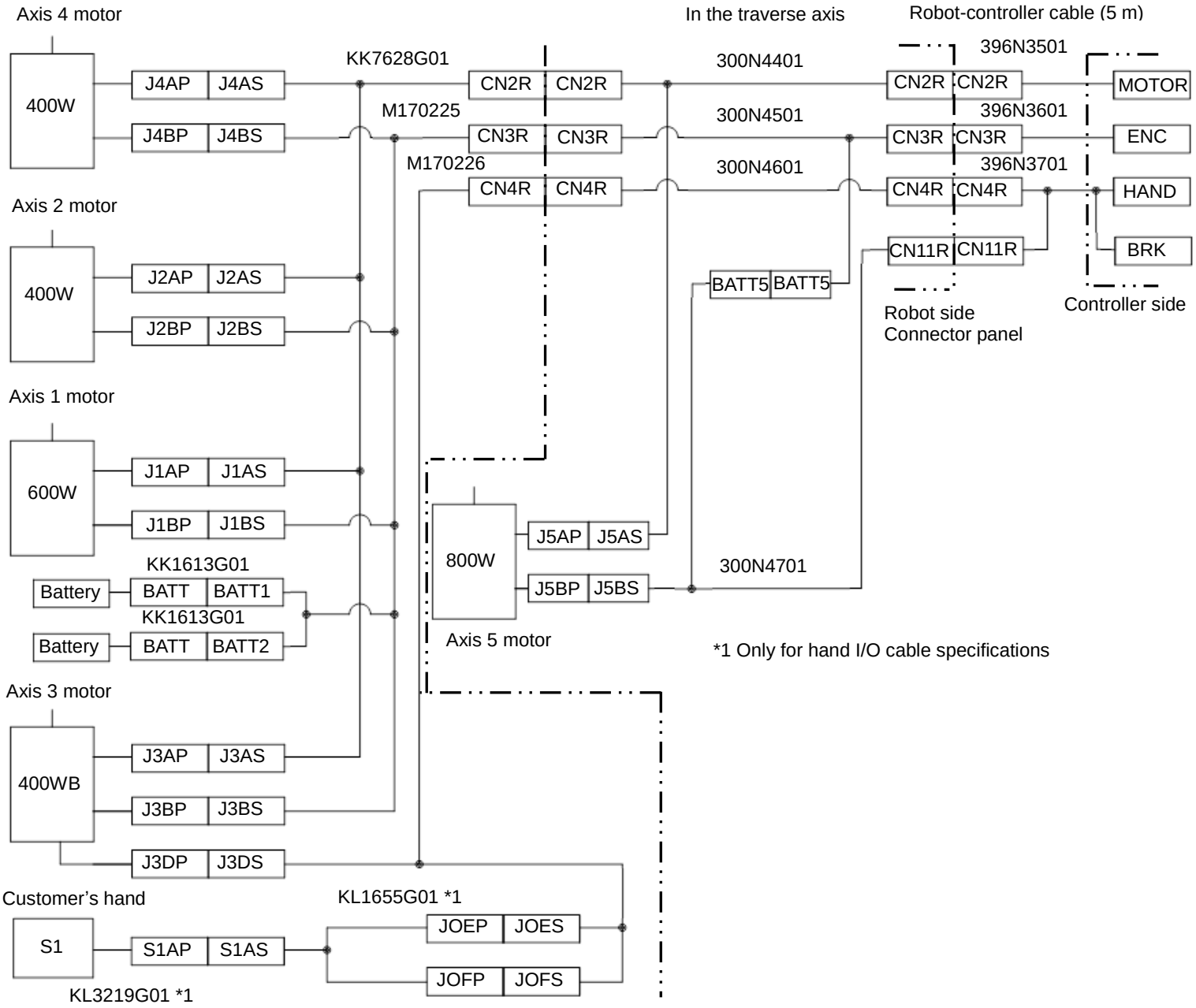


Fig. 5.3 Component Connection Diagram (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)